

BS Final Year Project

Handbook

Producing Quality FYP

3rd Edition

FAST School of Computing, Islamabad
National University of Computer and Emerging Sciences

2026

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The School of Computing extends its sincere appreciation to all faculty members whose dedication, expertise, and valuable contributions have helped shape and enhance the FYP Handbook.

Prof. Dr. Hasan Mujtaba
Head, School of Computing
NUCES-FAST, Islamabad
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1. Glossary

FYP – BS Final Year Project

Student – A student registered for an **FYP** at NUCES

FYP Group – A group of students formed as a team to work on an **FYP**

Faculty – Permanent Computing faculty at NUCES

Supervisor – A person responsible for the supervision of an **FYP Group**. S/he is a NUCES **Faculty** member.

Faculty Panel – A subset of **Faculty** responsible for the evaluation of an **FYP.** External experts of the relevant area can also be included.

Panel Head – A member of **Faculty Panel** responsible for the supervision of the panel examination

FYP Committee – A committee of **Faculty** responsible for executing the FYP process

FYP Coordinators – Responsible for managing **FYP** registrations, evaluations, and results

2. The Role of FYP Committee

For smooth execution of FYP process, the FYP committee carries out all the underlying supporting activities and provides the basic working platform for the execution of the FYPs.

The FYP committee works on different activities throughout an FYP lifecycle. Following is the detailed description of the roles and responsibilities of the FYP committee at different stages of the FYP process.

Step 1: Registration of FYP-1 Groups

The eligible students have to form FYP groups. As per current policy, an FYP group must consist of either 2 or 3 members. A group size of 3 members is preferable. After rigorous deliberations and keeping in view the previous experiences, the current policy does not allow a group size of more than 3 members and an individual student to separately register an FYP. The registration of an FYP group requires the students in the group to provide the following information to an FYP coordinator:

i. FYP title and a brief description of the FYP scope (one paragraph description of 8-10 key features)
ii. UN SDG(s) targeted by the FYP
iii. Name of the Supervisor and information of the group members (names, roll numbers, CGPAs)
iv. Category of the project and the technologies to be used in the project
v. Evidence of complex computing problem (refer to Appendix C for details related to complex computing problem)

The information is provided to the FYP committee using the online FYP group registration form. The submitted forms are then processed by the FYP committee. It checks the eligibility of the FYP-1 students and the projects. A list of registered FYP-1's is produced by the end of the second week of a semester.

Note: *The registration of an FYP group with the FYP coordinators is different from registering the FYP as a course on Flex. All students starting their FYPs must register first on Flex. In case an FYP is not registered on Flex, the student is debarred from registering FYP with the FYP committee.*

Step 2: FYP Proposal Defense

All members of an FYP group must appear before a faculty panel to present their FYP proposal. The FYP committee provides the format and duration of the presentations. For conducting the FYP proposal defense, the FYP committee performs the following tasks.

i. Formation of faculty panels
ii. Dissemination of the evaluation policy and guidelines to students and faculty
iii. Scheduling of FYP presentations
iv. Processing of the submitted forms and preparation of results
v. Dissemination of the faculty feedback to FYP groups and supervisors
vi. Re-evaluation of the rejected FYPs (with new/revised ideas and same/new supervisors/groups) with penalty of 5% of the newly secured marks

Step 3: FYP-1 Mid Evaluation

FYP groups submit the mid semester FYP-1 report (in soft form) to the FYP committee. They present their work to their supervisors. The development process of the FYPs follows iterative model. For this evaluation, the FYP committee shares the evaluation form with the supervisors.

Step 4: FYP Poster Preparation and its Evaluation

The FYP groups are required to create posters of their FYPs. The FYP committee is responsible for the following activities:

i. Collection of Posters from the FYP groups and verification of poster formats as per communicated guidelines. The posters are accepted only after the approval from the supervisors.
ii. Printing of the posters and their display in university (coordination with the university printer)
iii. Evaluation of the Posters
iv. Processing of the submitted forms and preparation of results

Note: *Poster evaluation may be merged (held in parallel) with the final FYP-1 evaluation.*

Step 5: Final FYP-1 Evaluation

The FYP groups present the work completed as part of FYP-1. The FYP-1 report is submitted to the FYP committee in soft form after the approval of the supervisors. The faculty panels grade the reports along with the presentation. The FYP report that is not approved by the supervisor may be rejected.

The FYP committee is responsible for:

i. Circulation of the presentation template and the evaluation guidelines to faculty and students
ii. Formation of the faculty panels and scheduling of the presentations
iii. Verification of the FYP reports
iv. Processing of the submitted forms and the preparation of results
v. Dissemination of the faculty feedback to groups and supervisors
vi. Preparation of the results (by applying an anomaly removal mechanism)

Step 6: Preparation of FYP-1 Grade

The FYP committee prepares the FYP-1 result. Every submission is verified and is accepted only after the approval of the supervisor. For grading, the standard university policy is followed. As per current policy, the **minimum passing marks for an FYP student are 50%.** In case of substantial difference of group members' performance reported/ verified by the FYP Supervisor, group members will be assigned grades based on their individual performances.

Step 7: Execution of FYP–2

FYP-2 is centered on the "job fair" (previously called Open-House), where companies are invited to view the FYPs and to hold the interviews. The samples of the forms used for FYP-2 are given in the forms section. The FYP handbook contains the tasks to be completed by the students before the job fair evaluation.

Step 8-a: Pre-Job Fair Evaluation (FYP-2 Midterm Evaluation)

A purpose of the pre-job fair evaluation is to determine whether the projects are ready to be presented to the industry professionals. The pre-job fair evaluation is held in the 8th week of FYP-2. Only the projects that clear the pre-job fair evaluation (or re-evaluation) are allowed to participate in the job fair. Projects that fail to appear in the job fair are penalized (up to 30% of FYP-2 marks).

The FYP Committee is responsible for:

i. Circulation of the presentation template and the evaluation guidelines to faculty and students.
ii. Formation of the faculty panels and scheduling of the presentations
iii. Approval of FYPs for appearing in the job fair
iv. Managing the project displays in the job fair

Step 8-b: Re-Evaluation of Rejected Projects

Full faculty panel (or multiple panels) evaluates the rejected FYPs for the final decision on acceptance/ rejection of the FYPs for the job fair.

Step 9: Managing the Job Fair

The job fair is usually held around 12th-14th week of FYP-2. All FYPs that pass pre-job fair evaluation(s) of appear in the job fair. The FYP committee coordinates with the department that is responsible for the industrial liaison for conducting the FYPs display related tasks in the job fair. It provides an important opportunity for the graduating students to appear before the prospective employers.

Step 10: Final FYP-2 Evaluation

The final FYP-2 evaluation is held in the 15th week of the semester. This evaluation is done by the supervisors. The FYP reports (submitted in soft form) are checked for plagiarism.

The FYP committee is responsible for:

i. Verification and vetting of the FYP reports
ii. Sharing the evaluation form with the supervisors
iii. Processing of the submitted forms and the preparation of results

Step 11: Final Deliverables and the Release of Grades

The FYP grades are released only upon receiving the final FYP report (in soft form) and other required documents (also in soft form). All reports must pass the plagiarism check specified by the university. The **Reports are accepted only after the approval from the supervisors**. In case of substantial difference of group members' performance reported/ verified by the FYP Supervisor, group members will be assigned grades based on their individual performances. The FYP committee is responsible for:

i. Processing of FYP-2 evaluations and grading
ii. Receiving of FYP-2 reports
iii. Vetting of FYP-2 reports
iv. Release of the grades based on the verification of the final deliverables

Similarly, the coordinators are also involved in any FYP-related purchase and act as the interface between the faculty and the IT department.

In summary, throughout FYP-1 and FYP-2, the FYP committee is responsible for:

i. Registration and maintenance of the records of FYP groups
ii. Setting of the FYP evaluation policies
iii. Scheduling and execution of the FYP evaluations
iv. Processing of the evaluation forms and the preparation of results
v. Conflict resolution among faculty members
vi. Dissemination of the results and the faculty feedback
vii. Verification of the FYP reports
viii. Coordination of the equipment purchase for FYPs
ix. Overseeing the FYP lab

The scope of the FYP committee does not cover:

1. Routine activities of monitoring and guiding the FYP groups (as it is a job of a supervisor). Similarly, the supervisors are responsible for tracking the weekly progress of the FYPs (on the prescribed forms).
2. Tracking of student attendance: This is a job of supervisors. Supervisors maintain the record of all meetings with students and provide the record to FYP coordinators at the end of a semester.
3. Providing support like computing and other facilities and monitoring student activities in the campus/labs. These are the jobs of the Labs and the department management.

3. The Role of a Supervisor

The role of a supervisor consists of:

1. Defining the scope of the project
2. Guiding the students about the technical issues related to the project
3. Monitoring the group's progress and assigning work to smoothly carry out the project
4. Making sure that the students regularly work on their project. A Supervisor should keep track of the weekly tasks completed by every student. This attendance/task allocation list should be submitted to the FYP committee at the end of a semester.
5. Making sure that all the deadlines set by the FYP Committee are met on time.
6. Reporting the non-functioning of a team to the FYP Committee, so that a timely action is taken to avoid the unnecessary wastage of time. **Students can be referred to the Head, FAST School of Computing for debarring from an FYP in case of failure to meet minimum attendance criteria**. An FYP group is required to visit its supervisor for an FYP meeting at least once in a week.
7. Training students for teamwork, professional ethics, presentations, etc. In particular, an FYP supervisor is responsible for ensuring that its group follows the iterative approach for the FYP.

4. Final Year Project Offering

The Final Year Project (FYP) is a compulsory requirement at undergraduate level. The students register FYP-1 in 7th semester and on successfully completing FYP-1, they register FYP-2 in 8th semester. The overall quality of the finished FYPs is a good representative of the quality of education imparted at the department. To ensure the high quality FYPs, a rigorous process is followed at NUCES.

FYP ideas can be given by the faculty, the students, or experts from the industry. The faculty should offer the FYPs well before the start of a semester. An FYP idea can be floated as one-page description in the following format:

1. Title of the project
2. Expected level of difficulty/ effort involved
3. General narrative description of the idea/ tasks — one or two paragraphs
4. References to a few resources such as books, book-chapters, papers, articles, technical reports, earlier project reports etc.
5. A list of expected deliverables or outcomes (if possible)
6. Any prerequisites or coursework for the project
7. Any special hardware requirement

These one-page descriptions are available for students to review. If they are interested in certain project, they need to study the referred materials and seek appointment from the concerned faculty. The supervisor and the students can then agree to carry out the project. **The FYP Committee should decide the FYP proposals after the FYP proposal defense, as per recommendations of the panel(s).**

5. The Process

The students are required to register the FYP-1 as a course on Flex. Once the students are registered on Flex, they become part of the FYP-1 process.

5.1 An Agile and Iterative FYP Process

The main goal is to move away from the lengthy requirements and the design process where there is no implementation in FYP-1 to a more balanced approach. The students are required to perform the complete cycles of requirements to analysis and design followed by the implementation in both FYP-1 and FYP-2. FYP-1 and FYP-2 are divided into iterations.

5.2 FYP Iteration

Iteration is a unit of work. FYP-1 and FYP-2 are now composed of a number of iterations. The size and duration of an iteration are left to the project supervisors.

In case of a development project, in every iteration, students pick a few use cases, expand and design them, followed by their implementation; before moving on to the next set of use cases. Students may use proxies and stubs where needed (for use case dependency). For every iteration, an FYP group should follow the standard analysis and design approach studied in the course work. Every iteration should result in a working product that can be demonstrated.

Students should ideally complete at least 2 iterations in FYP-1 (all combining for at least 60% implementation of the Project work). Remaining iterations are completed in FYP-2, thus completing the FYP project.

A sample execution of FYP-1 is given in Table 1a. FYP-2 follows the same pattern as given in Table 1b.

Table 1a: FYP-1 Evaluations

S. #	Deliverable/ Evaluation	Week #	Deliverable/ Presentation
1	Proposal Defense	4	Proposal document and presentation with POC-Lite
2	Mid Evaluation	8	FYP Report (version 1.0) - Project Vision - Research on related/ existing ideas/ products - Use Cases/ Use Case Diagram, SSD, SRS, Test Plan (Test Level, - Testing Techniques), Software Development Plan, Wireframes, UI Screens, DEI and Ethical/ Legal statements - Iteration – 1 **Design Phase** (Select the design that is appropriate for your project): - Structure Design: Domain Model/ Class Diagram, Component Diagram, Layer Diagram, Structure Chart - Behavior Design: Flow Diagram, Data Flow Diagram (DFD), Data Dictionary, Activity Diagram, Network Automata/ Graphs or State Machine, Call Graph or Sequence Diagram, Interaction Overview Diagram - For DB Representation: Schema Design/ ER Diagram - Data Structure and Algorithm Design **Development Phase:** - Comments, Naming Conventions, - Static Analysis of Code, etc., Unit Test Suites or Test Cases **Maintenance Phase:** - CI/ CD (e.g., SVN, GitHub, etc.) - Deployment Diagram - System-Level Test Suites, Test Cases (Validation and Testing) - Configuration/ Setup and Tool Manual Working Code + Demonstration
3	Poster Submission	13	Submission of Poster to the FYP Committee
4	Poster Evaluation	15	Posters are displayed in public area and are evaluated (can be done at the time of final evaluation)
5	Final Evaluation	15	- FYP Report (version 2.0) (with corrections) Adding subsequent iterations (after the 1st iteration) - Working Code + Demonstration - Maintenance phase (continued)

Any number of re-evaluations may be scheduled by the FYP Committee as per the recommendations of the panel(s).

Table 1b: FYP-2 Evaluations

S. #	Deliverable/ Evaluation	Week #	Deliverable/ Presentation
1	Pre – Job Fair Evaluation	8	Submission of the updated posters. Preparation of the demos. Submission of the updated report.
2	FYP Pre – Job Fair (Re-evaluation)	9	Re-evaluation of FYPs that fail first evaluation
3	Job Fair	12 – 14	FYP Projects are displayed to industry professionals. Students prepare and submit their CVs, FYP descriptions, pictures, and posters to be included in the Graduate Directory.
4	Final FYP-2 Evaluation	15	Complete FYP Report duly signed by the Supervisor adding: - Package & Deployment Diagrams - SVN, GitHub, etc. - Google Play or Apple Store (for mobile applications) - Configuration/ Setup and Tool Manual
5	FYP–2 Final Deliverables Submission	17	FYP report, FYP presentation, all codes, poster, any other documents etc. The report must be approved by the supervisor

FYP-1 Process

Step 1: Registration of FYP-1 Groups

During the semester break, faculty floats the proposals for the FYP through FYP coordinators. Students can also propose a project of their own but faculty should formally approve the project. The students who are completing their degree in the next semester are eligible to register FYP-1. **The Head, FAST School of Computing can allow relaxation of 1 course. Additional relaxation of a second course may be allowed by the campus Director.**

A **supervisor must be a full time faculty member** and may be assisted by an external supervisor. The students who are taking FYP from industry/ outside are required to have an internal supervisor from the faculty and they need to submit a three page (maximum) FYP proposal (whose requirements have been mentioned above) to the internal supervisor.

Once an FYP is registered, no change in FYP, group, or supervisor is allowed without the approval of the FYP committee.

Step 2: FYP Proposal Defense

The FYP-1 proposal defense is held within the first 4 weeks of the start of a semester and is used for vetting an FYP idea. All groups are allocated presentation times and they must

appear to present their FYP proposals. The FYP coordinators provide the format and duration of the presentations. The FYP committee create "Faculty Panels" to evaluate the FYPs.

The faculty panels take decisions related to the approval, re-evaluation, or rejection of the FYPs. The FYP groups may be asked to modify their FYP scope, etc. In extreme case, a faculty panel may even reject an FYP idea, requiring the students to re-register another FYP.

The FYP coordinators are responsible for gathering all the evaluation forms and preparing the results. For every group, the feedback provided by the faculty panel is shared with the groups and their supervisors. To preserve confidentiality, the comments are separately compiled and are forwarded without the names of the evaluators.

Note: Requirements for any additional hardware or software must be clearly indicated to the FYP committee. An application duly approved by the supervisor must be submitted, containing the information such as availability and cost of the hardware/ software resource. **Please note that the campus is not obliged to provide the required hardware/ software resource and such requests are considered on a case-to-case basis.**

Step 3: FYP-1 Mid Evaluation

The midterm evaluation is scheduled in the 8th week of a semester. The prime motivation behind this evaluation is to track the progress of the students and guide them in completing the remaining milestones. So, the primary focus of this evaluation is on the requirements and the design phase of FYP-1. FYP groups are required to submit the midterm report in soft form.

It is important to note that the number of use cases is not a measure for the quality of a project. The groups have to write as many use cases as it takes to properly document the requirements. For an FYP that lacks user interactions, this number may be quite low. However, if the students have only 2-4 use cases, they should be prepared to explain what they have been spending their time on? Perhaps, they have done some implementation, studied some new algorithms, done comparisons, etc. The groups should have working code to demonstrate the implemented iteration.

Step 4: FYP Poster Evaluation

The FYP Poster evaluation is held in the13th week of the semester by placing them on open display or at the time of the FYP-1 final evaluation.

Step 5: Final FYP-1 Evaluation

Till this stage, it is expected that the FYP groups should at least have the 60% implementation of their projects. Students are expected to complete the iteration-2 deliverable (the exact number of iterations to be completed in FYP-1 depends on the supervisor, but it is recommended that there should at least be 2 iterations).

Every group submits the FYP-1 report and delivers a presentation and demonstration of the work done publicly in front of a faculty panel, as per schedule announced by the FYP coordinators. The report should also include the plan of the work to be done in the next semester (for FYP-2). The detailed contents of the report are according to the requirements of an FYP and are decided by the supervisor and the group. The basic format and the guidelines for an FYP report is given at the end of this handbook. The report and the presentation are graded by the supervisor and the faculty panel**.**

Step 6. Preparation of FYP-1 Grade

The FYP coordinators prepare the FYP-1 result. The grades are released only after the submission of FYP-1 report (in soft form). Every submission is verified and it is accepted only after the approval of the supervisor. The late submissions are liable to get penalty decided by the FYP committee. The students may get a zero for a late submission.

Note

- Exact due dates are announced at the start of a semester.
- Supervisor/ teams should have their internal deadlines in order to meet the external deadlines.
- All groups must follow the iterative approach

● FYP-2 Process

1. Those students who have passed FYP-1 register for the FYP-2 course on Flex and continue with their allotted FYPs.
2. The academic office provides the official list of registered students (in FYP-2) to the FYP committee.
3. The mid semester (pre-Job Fair) evaluation is held in the 8th week. Each group gives a demonstration of its FYP. The evaluation is done by the supervisor and the faculty panel.
4. The preparation for the job fair should be kept in mind as it is the earliest stage when recruitment process is initiated and it is pragmatic to impress/ attract the companies by showcasing the students' capabilities. So, a working system should be ready by the 12th week. The FYP coordinators will make a notification in this regard giving a complete schedule of the event.
5. The final evaluation is held in the 15th week. A comprehensive FYP report (in soft form) is submitted before the presentation.
6. Each group submits the final FYP report (in soft form) and all other deliverables (also in soft form) before the 18th week to the FYP committee. The submitted report and deliverables should include the corrections and amendments proposed during the final evaluation. The final report should follow the format specified by the university.
7. Final Deliverables
   i. **Report** (soft copy of the final report and the complete power point FYP presentation)
   ii. **Reference** (reference material like articles, tool information etc.)
   iii. **Code** (complete source code of the project)
   iv. **Demonstration** (the executable in working order and a readme file containing the information about the software requirements (tools) and hardware requirements for the FYP as well as the instructions or the steps (Soft copy of the User manual) for running the FYP executable).

Note

1. Exact due dates are announced at the start of a semester.
2. Supervisor/ teams should have their internal deadlines in order to meet the external deadlines

6. Evaluation

A recommended guideline for the composition of the grade is as follows:

Evaluators' Weights for both FYP-1 and FYP-2

Evaluation	Weight
Supervisor(s)	≈ 35-60 %
Faculty Panel (the entire panel is treated as examiner)	≈ 40-65 %

Refer to appendices A and B to see the evaluation criteria and rubrics for different evaluation stages.

Compilation of Results

The evaluations submitted by a supervisor and the faculty panel are compiled and converted into a letter grade by the FYP Coordinators.

- The recommended share of the Supervisor and the Faculty Panel may vary from campus to campus as per the range given in the above table.
- The letter grade is assigned after taking into account any deduction for the late submission of proposals, reports, failure in any evaluation etc.

Miscellaneous Rules

The entire FYP process is designed to ensure that students are able to develop and deliver high quality FYPs. At the same time, the university has put in place rigorous rules to ensure fairness of the FYP process. Following additional rules apply:

1. Failing the FYP-1 proposal defense requires the group to re-submit a new/improved proposal. This can be done with the same supervisor or new supervisor.
2. Once an FYP idea has been registered with a given supervisor, the change of supervisor can only be done with the consent of the existing supervisor. Another option is to change the idea entirely and register with a new supervisor, giving up the previous idea.
3. FYP-1 and FYP-2 must be registered consecutively. There cannot be a frozen semester in between FYP-1 and FYP-2.
4. In case of receiving an "F" in FYP-1, the students must re-register the FYP (same process as FYP-1). However, if they wish to change supervisor, then they must obtain permission from the supervisor through the *supervisor change form*. Another option is to drop the idea and register a new idea with a new supervisor. This only requires approval from the FYP Coordinators.
5. In case of receiving an "F" in FYP-2, the students cannot change supervisor without the consent of the existing supervisor. FYP-2 must be re-registered as per normal practice. However, the idea cannot be changed at this stage as this may require re-doing the FYP-1 (FYP-1 and FYP-2 must be on the same idea). The Head, FAST School of Computing in consultation with the FYP Coordinators may grant an exemption to this rule. The change of supervisor is allowed, but only through the explicit approval of the supervisor and the FYP coordinators.

6. In case the supervisor leaves the university, the FYP committee shall assign a new supervisor.
7. If case of the late submission of the final report (or submitting a report without supervisor's approval), the awarded grade may be converted to an "F", in line with the university policy.
8. Any student may be referred to the Head, FAST School of Computing or the campus Director for missing the scheduled appointments with the Supervisor. Such students may be debarred from the final evaluation and be awarded an "F" grade on the recommendation of the supervisor and approval from the Head, FAST School of Computing. However, the supervisor must inform the FYP committee that the student is not coming to meetings and must issue a formal warning to the student.

7. Guidelines for the Preparation of FYP Poster

The FYP poster is a condensed representation of an FYP. It should be neat, attractive, and very inviting. Posters are evaluated by the faculty panel. The following is a list of recommended poster guidelines:

- Poster size must be 22.5" x 34.5".
- The orientation of the poster ***should be Portrait***.
- Don't use too much text - just highlight your major points. Use bullets whenever possible.
- Make sure that the font is large enough for people to see it from a meter away.
- Illustrations and visualizations of concepts (maps, pictures, photos, design drawings, diagrams, tables, charts, graphs, "screen captures") look nice and can often say more than words.
- Make sure you spell check! (Also, note that capitalized words are often skipped by most spell checkers.)
- The top-left part may contain the title of the project, names of the group members and the Supervisor(s).
- The top-right part may contain the Project Goals.
- The bottom part may contain the Project Plan and the list of tentative deliverables for the final FYP-1 evaluation.
- While staying within these guidelines, there is a plenty of scope for an individual creativity through the use of different fonts, colors, backgrounds and graphics.
- Do not include too much information. Too much text may obscure the main message of the poster. Limit the information to the key information; rely on answering questions and on your project summary to get across the details.
- Use graphics where appropriate.
- The original Photoshop/ Corel file of the poster should be submitted along with a jpeg image of the poster.
- Recommended tools for making poster are Adobe Photoshop or Corel Draw. If a group uses some other tool, he/she will be responsible for the printing of Poster.

8. Guidelines for FYP Report Format

Title Page

The title page should include the title of the report along with the name(s) of the department or university/organization for which the report is written and the year of submission. Also included on the title page should be the name(s) of the author(s) of the report along with the name (s) of the supervisor (s). Title Page is followed by a blank page. A sample is shown below.

Final Year Project

Handbook (Title)

Producing Quality FYP (Subtitle)

FYP Team

Mr. Hassan Mustafa

Mr. Shoaib Mehboob

Supervised by

Mr. Shams Farooq

FAST School of Computing

National University of Computer and Emerging Sciences

[Campus], Pakistan

2026

Students' Submission

This includes the title of the report and its occasion.

Anti-Plagiarism Declaration

This is to declare that the above FYP report produced under the:

Title: _____

is the sole contribution of the author(s) and no part hereof has been reproduced on **as it is** basis (cut and paste) which can be considered as **Plagiarism.** All referenced parts have been used to argue the idea and have been cited properly. I/We will be responsible and liable for any consequence if violation of this declaration is determined.

Date: _____

Student 1
Name: _____

Signature:_____

Student 2
Name: _____

Signature:_____

Student 3
Name: _____

Signature:_____

Supervisor (Faculty)
Name: _____

Signature:_____

Authors' Declaration

This states Authors' declaration that the work presented in the report is their own, and has not been submitted/presented previously to any other institution or organization.

Abstract (optional) (50 to 125 words)

Three to five sentences describing the essence of the work. An abstract is a short, 50-125 words summary of a work. An Abstract should state the purpose, findings, and conclusions of your work without commenting on or evaluating the work itself. Put the abstract on a separate page that follows the title page.

Acknowledgments (if any)

Executive Summary:

The executive summary should be one to two pages' overview of the information contained in the FYP report. It should give the reader an easy reference, in a very brief form, to the important information contained in the report and explained in more detail in the body of the report. People reading the report will use this section as a reference during presentations.

Table of Contents

The table of contents lists the information contained in the report in the order in which it will be found. All major topics of interest should be listed.

Introduction

The introduction should contain a brief overview of the problem being addressed and the background information needed for the reader to understand the work being done and the reasoning behind it. After reading the introduction, the reader should know exactly what the report is about, why the work was done, and how this work adds to the knowledge that the reader may have about the topic.

Body Chapters

Conclusions and/ or Recommendations

Present a summary of what you found in the results section. Here you should describe the techniques that you used for each analysis and the results of each analysis.

Conclusions are broad generalizations that focus on addressing the questions for which the project was conducted. Recommendations are your choices for strategies or tactics based on the conclusions that you have drawn. Quite often authors are tempted to speculate on outcomes that cannot be supported by the findings. Do not draw any conclusions or make any recommendations that your work cannot clearly support.

References

This section should be a listing of all existing information sources used in the FYP. It is important to allow the reader to see all of the sources used and enable the reader to further explore those sources to verify the information presented.

Appendix/ces

This section should include all supporting information from the project that was not included in the body of the report. You should include surveys, complex statistical

calculations, certain detailed tables, and other such information in an appendix. The information presented in this section is important to support the work presented in the body of the report but would make it more difficult to read and understand if presented within the body of the report. Appendixes may include maps, graphs, charts, or other helpful material.

Cite the appendix items in the report narrative (write "see Appendix A") and organize appendices (e.g., Appendix A, Appendix B, etc.)

Any tables, figures, form, or other materials that are not totally central to the analysis but that need to be included are placed in the Appendix.

Footnotes/ End Notes

Footnotes are located at the bottom of a page. End notes are like footnotes but are located at the back rather than the bottom of each page. These would include all of the references for all works cited in the review of related literature or any other sections of the report as well as the references for quotations, either direct or indirect, taken from other sources, or any footnote comments that might have been included. These are listed in numeric order as presented in the text.

1. Lani Arredondo (1999) *Business Presentations*, McGraw-Hill, pp. 23-34.
2. Jann Schill (1996) *On Purpose,* Heinemann, Australia, p. 23.
3. Paul Tench (1981) *Pronunciation Skills*, Macmillan, p. 23.

In case an endnote is followed by another citing the same source, only the word: “Ibid” (as above) shall substitute for the entire text. If the page no. is different then the page number shall appear after the word: “Ibid”. For example:

1. Jann Schill (1996) *On Purpose,* Heinemann, Australia, p. 23.
   Ibid., p. 54.

In case the same source comes to be mentioned after a few interrupting sources then the word “op cit” (mentioned earlier is written), followed by the page number.

Bibliography or Literature Cited

Bibliography will appear in alphabetic order of the authors’ surnames. Author’s surname shall come first as is the case in the catalogues of libraries.

Arredondo, Lani (1999) *Business Presentations*, McGraw-Hill.
Schill, Jann (1996) *On Purpose,* Heinemann, Australia.
Tench, Paul (1981) *Pronunciation Skills*, Macmillan.

Note: Use underlining (italics) for titles of books, periodicals newspapers, works of art, ships, etc. Articles or chapters in books, periodicals, or journals shall come in double quotes.

Paper

Standard A4 size
Width: 8.27"
Height: 11.69"
Weight: 90 Grams

Fonts, Type Styles

Font Size = 11 (Normal Text)
Font = Times New Roman
Title= 26 bold (Times New Roman)
Sub-title=20 bold (Times New Roman)
Heading 1 (Font Size) = 16 (Bold)
Font = Arial
Heading 2 (Font Size) = 14 (Bold)
Font = Arial
Heading 3 (Font Size) = 13 (Bold, Italics)
Font = Arial

Margins

Top = 1.5"
Bottom = 1.0"
Left = 2.0"
Right = 1.0"

Spacing

Line Spacing = 1.5
Paragraph Spacing = 6 pts

Indentation

Indent all quotations comprising 4 or more lines by 5 spaces from left.

Page Numbers

Except for the title page, number all pages which come before the first page of the body chapters consecutively with lower case roman numerals (i, ii, iii, iv,…).

The first page with Arabic numeral (1, 2, 3, and so on) starts from the page of the introduction but it is mentioned on page 2 onwards.

Mention page numbers on the top right of the page. The first page of each section or chapter will not carry the page number; however, the page number will be counted for the proceeding page.

Headers

The header will comprise the title of the Project report. On every odd page will appear the title of the report while on the even pages the title of the chapter or section will be mentioned. The first page of every section or chapter shall not carry the header.

9. FYP Report Contents

The basic contents for report are given below as a guideline. A specific and tailored format according to an FYP's need is suggested by a supervisor.

1. Introduction
2. Research on existing ideas/ products
3. Project Vision
    - 3.1. Problem Statement
    - 3.2. Business Opportunity
    - 3.3. Objectives
    - 3.4. Project Scope
    - 3.5. Constraints
    - 3.6. Stakeholders Description
    - 3.6.1. Stakeholders Summary
    - 3.6.2. Key High Level Goals and Problems of Stakeholders
    - 3.7. DEI and Ethical/ Legal statements
4. Software Requirement Specifications
    - 4.1. List of Features
    - 4.2. Functional Requirements
    - 4.3. Non-Functional Requirements
5. Iteration Plan
6. Iteration 1 (all artifacts mentioned in Table 1a)
7. Iteration 2 (all artifacts mentioned in Table 1a)

    and so on … (for subsequent iterations)

8. Implementation Details (not the programming code but the algorithmic and procedural details especially related to the hidden/ backend algorithms that are not covered in the design)
9. User Manual

References

Appendices

FYP Tracking/Monitoring Form
(For Supervisors)

FYP Title:_____

Students:_____ **Semester:** _____

Period from: _____ **to:** _____

Date	Work Completed	Work Assigned	Target Date

Supervisor: _____ **Signature:** _____

Appendix A: Iteration Intent Statements

This document defines the purpose, evaluator focus, quality indicators, and developmental intent of each of the five FYP evaluation stages.

The Five Evaluation Stages

FYP-1 — 100 Marks Total

Stage	Title	Evaluator	Semester / Week	Duration	Marks	Doc. Sub-weight	Primary Purpose
0	Proposal Defense	Panel + Supervisor	FYP-1 / Week 4	20 min.	30 marks (30%)	—	Scoping and feasibility gate — validate CCP, approach, and team capacity
1	FYP-1 Mid	Supervisor only	FYP-1 / Week 8	15 min.	20 marks (20%)	5 marks (25% of 20)	Formative developmental checkpoint — design reasoning and early momentum
2	FYP-1 Final	Panel/ Supervisor (demo) + Supervisor/ examiner (report)	FYP-1 / Final week	15 min. + report	50 marks (50%)	7.5 marks (15% of 50)	Integration readiness gate — quality, comprehension, and structural soundness, poster evaluation

FYP-2 — 100 Marks Total

Stage	Title	Evaluator	Semester / Week	Duration	Marks	Doc. Sub-weight	Primary Purpose
3	FYP-2 Mid / Pre-Job Fair	Panel + Supervisor	FYP-2 / Week 8	15 min.	50 marks (50%) ★	—	Public readiness assessment — integrated system and job fair eligibility decision
4	FYP-2 Final	Supervisor only	FYP-2 / Final week	15 min. + report	50 marks (50%)	7.5 marks (15% of 50)	Closure and contribution evaluation — honest account, documentation, lifecycle reflection

★ *FYP-2 Mid carries equal weight to FYP-2 Final but serves the additional role of job fair eligibility gate. Documentation sub-weight of 15% applies to both Finals only: 42.5 marks demonstration/ defense + 7.5 marks report. Fractional marks resolved at grade entry per institutional policy.*

Documentation Weight Breakdown (Finals Only)

Component	FYP-1 Final (50 marks)	FYP-2 Final (50 marks)
Live Demonstration + Defense (85%)	42.5 marks	42.5 marks
Report / Documentation (15%)	7.5 marks	7.5 marks
Total	50 marks	50 marks

Stage 0 **Proposal Defense** *FYP-1, Week 4*	*Evaluator* **Panel + Supervisor** *Duration* **20 minutes** *Marks* **30 marks (30% of FYP-1)** *Scoping gate*

PURPOSE

The proposal defense is a scoping and feasibility gate, not an idea competition. Its purpose is to confirm that the group has identified a problem of sufficient complexity to qualify as a Complex Computing Problem (CCP), that they have surveyed the landscape of existing solutions honestly, and that their proposed approach is technically plausible within the time and team constraints of a two-semester project. The POC-lite is required not as evidence of completion but as evidence of seriousness — that the group has moved beyond theoretical discussion and made contact with the actual technical reality of their idea. The vibe coding or AI-assisted SDLC framework the group intends to follow should be named and briefly justified at this stage.

Primary question the evaluator is answering

> *"Is this problem real, is this approach viable, and is this group capable of executing it over two semesters?"*

What good looks like at this stage

● A clearly articulated gap between existing solutions and the proposed contribution, located at a level of difficulty that requires in-depth computing knowledge to address — not a feature addition or a re-implementation of a known system.
● A technical approach — architecture, technology choices, data or model strategy — that is justified rather than assumed; the group can explain why they chose this approach over alternatives they considered.
● A POC-lite that reveals something meaningful about a risk or feasibility question: an API integration check, a baseline run, a data pipeline sample, or a functional prototype interaction — not only a UI mockup or a slide with a system diagram.
● A work division specific enough that each member has an identifiable ownership stake from day one, with milestones balanced across both frontend and backend or research and implementation concerns.
● Explicit disclosure of which AI tools, APIs, or pretrained models are expected to be used, and at what level of involvement — support, core-assist, or core-replacement.

The key developmental shift

→	After the proposal defense, the group moves from problem identification to structured design and early implementation. The panel sets the first milestone register at this stage, which becomes the accountability backbone for FYP-1 Mid. Students who leave the proposal defense with vague milestones will struggle to demonstrate meaningful progress at FYP-1 Mid.

Stage 1 **FYP-1 Mid** *FYP-1 · Week 8*	*Evaluator* **Supervisor only** *Duration* **15 minutes** *Marks* **20 marks (20% of FYP-1) — including 5 marks documentation** *Formative checkpoint — supervisor only*

PURPOSE

The FYP-1 Mid is a formative supervisory checkpoint, not a panel gate. Its purpose is to give the supervisor a structured opportunity to assess whether the group's design thinking is maturing, whether implementation momentum is real, and whether the panel's proposal-stage milestones have been meaningfully addressed. Because this evaluation is conducted by the supervisor alone, it is deliberately diagnostic and developmental in tone. Students should feel safe surfacing difficulties here — blocked integrations, scope concerns, AI-generated outputs they do not fully understand — because this is the last private opportunity before the panel re-engages at FYP-1 Final. The supervisor's role here is part evaluator, part coach.

Primary question the evaluator is answering

> *"Is the group on a trajectory that will produce something defensible and substantively understood by the end of FYP-1, and what must change in the remaining weeks to ensure that?"*

What good looks like at this stage

- A live module or prototype that goes beyond the POC-lite — something that implements actual project logic, not just interface scaffolding or a re-demonstration of the proposal spike.
- Design artefacts that have evolved since the proposal: they reflect decisions made during implementation, not just the original plan. Changes should be visible and briefly explained.
- The group can walk the supervisor through their codebase at the file and function level, demonstrating that vibe-coded or AI-assisted output has been reviewed, understood, and genuinely owned — not simply accepted and submitted.
- The milestone log is up to date and honest about delays. Where a milestone was missed, a recovery plan with revised owners and dates is presented.
- At least one module has been validated beyond the happy path — a failure case, an edge condition, or a boundary test has been documented and discussed.

The key developmental shift

→	After FYP-1 Mid, the group has roughly half a semester to reach a state where a panel can evaluate significant implemented work. The supervisor issues targeted corrections through the milestone register, focusing the group's remaining effort on the areas most likely to produce defensible, integrated work by FYP-1 Final. This is also the point at which scope adjustments should be formally discussed, before they become crises at the end of semester.

Stage 2 **FYP-1 Final** *FYP-1 · Final week*	*Evaluator* **Panel + Supervisor (demo) + Supervisor/ examiner (report review)** *Duration* **15 min. presentation + report review (by Supervisor/ examiner)** *Marks* **50 marks (50% of FYP-1) — including 7.5 marks documentation** *Integration readiness gate*

PURPOSE

The FYP-1 Final is an integration readiness gate. Its purpose is not to confirm that the project is complete — that is not expected at this stage — but to confirm that what has been built is coherent, genuinely understood, and structurally sound enough to serve as the foundation for full system integration in FYP-2. The panel evaluates the presentation; the supervisor reviews the report. These are two lenses on the same underlying reality: the presentation reveals live understanding and ownership, while the report reveals depth of design reasoning and documentation discipline. Both must be consistent with what was actually demonstrated.

Primary question the evaluator is answering

> *"Is the work completed in FYP-1 of sufficient quality, comprehension, and structural integrity to support full integration and evaluation in FYP-2?"*

What good looks like at this stage

- At least 60% of committed scope is implemented in a way that is modular, readable, and buildable without the original developer present — not just running on one laptop with undocumented dependencies.
- Each member can explain their owned module — its design decisions, its dependencies, the edge cases it handles — without deferring to the AI tool that helped generate it. The panel may ask any member to show any part of the codebase.
- Testing is not deferred: completed modules have been validated beyond the happy path, with at least some failure cases documented and discussed rather than hidden.
- The report reflects the actual implemented system, not the intended one — requirements, design artefacts, and implementation documentation are consistent with what was demonstrated live.
- The FYP-2 evaluation plan is articulated: what baselines, benchmarks, or user scenarios will be used to assess whether the system actually solves the CCP.

The key developmental shift

→	After FYP-1 Final, the focus shifts entirely from individual module development to system integration, evaluation design, and CCP resolution. The panel sets the FYP-2 milestone register at this stage. Students who enter FYP-2 with well-understood, modular code will integrate smoothly; those who enter with AI-generated code they cannot explain will find FYP-2 significantly harder. The FYP-1 Final report is the contract for FYP-2.

Stage 3 **FYP-2 Mid / Pre-Job Fair** *FYP-2 · Week 8*	*Evaluator* **Panel + Supervisor** *Duration* **15 minutes** *Marks* **50 marks (50% of FYP-2) ★ Job Fair Gate** *Highest weight — dual academic and public gate*

PURPOSE

The FYP-2 Mid is the most consequential evaluation in the entire FYP process. It carries the highest single-stage impact and serves a dual purpose simultaneously: it is an academic milestone assessment and a public readiness gate for job fair participation. These two purposes are related but distinct and the panel must assess both explicitly. Academic readiness asks whether the integrated system demonstrates coherent engineering work that addresses the Complex Computing Problem. Public readiness asks whether the system is stable, presentable, and honest enough to represent the institution to industry partners. A project can pass one dimension and fall short on the other — the panel decision must address both.

Primary question the evaluator is answering

> *"Has the group produced an integrated, working system that demonstrates real engineering ownership and is ready to be presented publicly to industry — and if not, what must change before the job fair?"*

What good looks like at this stage

● A fully integrated system where all major modules are connected and data and control flow across them is coherent — not a collection of independently-working components that have not yet been joined.
● The system handles realistic scenarios, not carefully constructed demo paths only. It fails gracefully when pushed and the group can discuss known limitations honestly rather than deflecting.
● Each member can demonstrate and explain their owned modules in front of an industry audience — not only in a technical panel setting. Communication is clear enough for a non-specialist evaluator at the job fair.
● The Complex Computing Problem is visibly addressed in the system — not only described in slides. The panel can see where the non-trivial technical work lives and probe it.
● Testing and evaluation evidence is present: integration tests, known bugs with mitigation plans, and at least preliminary comparison against the baseline or benchmark defined in FYP-1 Final.
● Tool and GenAI use is fully disclosed and defensible — every member can explain what the AI contributed and what they added, modified, or validated beyond it.

The key developmental shift

→	After FYP-2 Mid, the project enters its final phase. Groups recommended for the job fair represent the institution publicly and should treat industry feedback as input for final touches. Groups not recommended receive a structured re-evaluation timeline with specific conditions stated in the panel action register. In both cases, the FYP-2 Final with the supervisor is the closure event — not another high-stakes gate, but a professional handover conversation.

Stage 4 **FYP-2 Final** *FYP-2 · Final week*	*Evaluator* **Supervisor only** *Duration* **15 minutes + final report review** *Marks* **50 marks (50% of FYP-2) — including 7.5 marks documentation** *Closure and reflection — supervisor only*

PURPOSE

The FYP-2 Final is a closure and contribution evaluation, not a repetition of the job fair or the FYP-2 Mid panel. Because the significant summative judgment has already been made at FYP-2 Mid, this evaluation serves a different and more reflective function: it is a professional handover conversation between the group and their supervisor, confirming that the system is complete and honestly documented, that the Complex Computing Problem has been addressed and its resolution evaluated against substantiated evidence, and that each individual can give a genuine account of their contribution across the full two-semester lifecycle.

Primary question the evaluator is answering

> *"Can each student give an honest, substantiated account of what the system does, what it does not do, what they personally built and understood, and what they would do differently — and does the final report reflect that account faithfully?"*

What good looks like at this stage

- The system works as demonstrated at FYP-2 Mid, with final touches and any job-fair-identified improvements incorporated. No regression from the mid evaluation state.
- The final report is a faithful document of the delivered system — not aspirational, not padded with planned features never implemented — with clear traceability from requirements through design through implementation through evaluation results.
- Each member gives a reflective individual account: what they owned, what the AI tools contributed and what they added beyond the tools, what the evaluation revealed about the system's limitations, and what they would approach differently.
- The limitations section of the report is substantive and honest — it identifies real gaps in functionality, evaluation coverage, or CCP resolution, not generic statements about future work.
- The supervisor is looking for genuine understanding and professional maturity, not a polished final performance. A student who can articulate clearly what they did not fully understand is demonstrating more than one who claims complete mastery.

The key developmental shift

→	This is the end of the FYP journey. A student who exits the FYP-2 Final with a clear, honest, and substantiated account of a real computing contribution is prepared to enter professional practice and to engage in the life-long learning the Seoul Accord expects. The final report, once accepted by the supervisor, becomes the permanent academic record of the project.

Appendix B: FYP Evaluation Forms and Rubrics

Guiding principle: every evaluation item must ask what evidence proves that the students — not only tools — understood, designed, built, validated and can modify the project.

Each criterion is mapped to a single primary CLO and Seoul Accord GA for attainment reporting.

1. Seoul Accord Graduate Attributes — Computing Professional (4-Year BS)

The following ten Graduate Attributes are defined by the Seoul Accord (Section D) for the Computing Professional graduate. GA1 is a program-level attribute and is not directly mapped to individual CLOs. All other GAs are covered by the FYP CLO set below.

Source: Seoul Accord Section D (seoulaccord.org/document.php?id=79)

GA	Attribute Name	Computing Professional Descriptor
GA1	**Academic Education**	Completion of an accredited program designed to prepare graduates as computing professionals. (Program-level attribute — not mapped to individual CLOs.)
GA2	**Knowledge for Solving Computing Problems**	Apply knowledge of computing fundamentals, computing specialization, and mathematics/science/domain knowledge to the abstraction and conceptualization of computing models from defined problems and requirements.
GA3	**Problem Analysis**	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
GA4	**Design / Development of Solutions**	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
GA5	**Modern Tool Usage**	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
GA6	**Individual and Team Work**	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
GA7	**Communication**	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
GA8	**Computing Professionalism and Society**	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
GA9	**Ethics**	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.
GA10	**Life-long Learning**	Recognize the need, and have the ability to engage in independent learning for continual development as a computing professional.

1a. CLO-to-Graduate-Attribute Mapping

Each CLO is mapped to a single primary GA for attainment reporting, with secondary GAs noted. This version promotes five GAs from secondary-only to primary, ensuring all Seoul Accord GAs (except GA1) have at least one primary CLO home.

CLO	CLO Summary	Primary GA	Justification for Primary Mapping	Secondary GA(s)
FYP1-CLO1	Defend idea, identify CCP via literature and existing solutions	**GA3** *Problem Analysis*	Researching literature, formulating and identifying complex computing problem maps directly to GA3. No change — unambiguous primary match.	GA2 — computing knowledge applied to conceptualize the problem
FYP1-CLO2	Define requirements, scope, feasibility, risks, iteration plan, ethics/ DEI/ legal issues	**GA8** *Computing Professionalism and Society*	CLO2 explicitly requires DEI, legal and ethical issues, risk register, and professional responsibility in planning — the strongest expression of GA8 in the curriculum. Promoted from secondary.	GA4 — requirements as opening phase of design; GA9 — ethical responsibilities
FYP1-CLO3	Design system using UML/ architecture/ data/ process artefacts, justify choices	**GA4** *Design / Development of Solutions*	Producing and justifying architectural and design artefacts for a complex problem is the clearest remaining primary expression of GA4.	GA2 — computing knowledge required to make justified design decisions
FYP1-CLO4	Implement initial prototype/ module, validate using tests/ baselines	**GA2** *Knowledge for Solving Computing Problems*	Implementing a working prototype requires the deepest application of computing fundamentals. GA2 defined as applying knowledge to abstraction and conceptualization of computing models. Promoted from secondary.	GA4 — implementation as part of the design-and-evaluate cycle
FYP1-CLO5	Justify tool/ API/ GenAI use, demonstrate human review and modification	**GA5** *Modern Tool Usage*	Creating, selecting, adapting and applying tools with understanding of limitations is the precise definition of GA5. Direct and unambiguous match.	— (direct match; no meaningful secondary)
FYP1-CLO6	Present and defend project as team, traceable individual contribution	**GA6** *Individual and Team Work*	CLO6 evidence — repo history, milestone log, member ownership — is fundamentally about individual and team contribution. Promoted from secondary to give GA6 its only primary home.	GA7 — presentation and Q&A defense as communication evidence
FYP2-CLO1	Refine scope, design, requirements based on FYP-1 feedback and change requests	**GA10** *Life-long Learning*	Acting on structured critique and adapting professional practice across semesters is the most direct expression of GA10. Promoted from secondary to give GA10 its only primary home.	GA4 — iterative refinement of design artefacts
FYP2-CLO2	Implement and integrate modules into coherent system addressing CCP	**GA4** *Design / Development of Solutions*	System integration that resolves a complex computing problem is the culmination of the GA4 design-and-evaluate cycle. Strongest remaining GA4 primary home.	GA3 — solving the CCP to substantiated conclusions through integration
FYP2-CLO3	Apply systematic testing and validation techniques	**GA5** *Modern Tool Usage*	Selecting and applying appropriate testing and validation techniques with understanding of their limitations aligns directly with GA5.	GA4 — testing as evaluation of the designed solution
FYP2-CLO4	Evaluate system against baselines, benchmarks, user scenarios, success criteria	**GA3** *Problem Analysis*	Evaluating against baselines and benchmarks requires reaching substantiated conclusions — the precise language of GA3. Promoted to give GA3 a primary home in FYP-2.	GA4 — evaluation as the second half of the design-and-evaluate cycle
FYP2-CLO5	Responsible practice: explain tool role, limitations, risks, human contribution	**GA9** *Ethics*	CLO5 is about responsible engineering practice — disclosure obligations and accountability. GA9 defined as understanding and committing to professional ethics and norms. Promoted to give GA9 its only primary home.	GA5 — tool usage with understanding of limitations; GA8 — professional responsibilities

CLO	CLO Summary	Primary GA	Justification for Primary Mapping	Secondary GA(s)
FYP2-CLO6	Final demo, technical defense, verifiable individual contribution across lifecycle	**GA7** *Communication*	Final presentation, live demo, technical defense and written report constitute the complete GA7 communication evidence set across the full lifecycle.	GA6 — verifiable individual contribution demonstrates effective team membership

GA Coverage Summary

✦*= Promoted from secondary-only to primary in this revision. GA4 reduced from six primary mappings to two, redistributing coverage to previously unanchored GAs.*

GA	Attribute Name	FYP-1 CLOs (Primary)	FYP-2 CLOs (Primary)	Also appears as Secondary in	Coverage
GA1	**Academic Education**	—	—	—	Program-level only
GA2	**Knowledge for Solving Computing Problems**	FYP1-CLO4 ✦	—	FYP1-CLO1, FYP1-CLO3	✓Primary (FYP-1)
GA3	**Problem Analysis**	FYP1-CLO1	FYP2-CLO4 ✦	FYP2-CLO2	✓Both semesters
GA4	**Design / Development of Solutions**	FYP1-CLO3	FYP2-CLO2	FYP1-CLO2, FYP1-CLO4, FYP2-CLO1, FYP2-CLO3, FYP2-CLO4	✓Both semesters
GA5	**Modern Tool Usage**	FYP1-CLO5	FYP2-CLO3	FYP2-CLO5	✓Both semesters
GA6	**Individual and Team Work**	FYP1-CLO6 ✦	—	FYP2-CLO6	✓Primary (FYP-1)
GA7	**Communication**	—	FYP2-CLO6	FYP1-CLO6	✓Primary (FYP-2)
GA8	**Computing Professionalism and Society**	FYP1-CLO2 ✦	—	FYP2-CLO5	✓Primary (FYP-1)
GA9	**Ethics**	—	FYP2-CLO5 ✦	FYP1-CLO2	✓Primary (FYP-2)
GA10	**Life-long Learning**	—	FYP2-CLO1 ✦	—	✓Primary (FYP-2)

1b. Recommended CLO Structure and Evidence Base

CLOs are written for accreditation mapping and future-proofing. GenAI and external tool use are kept visible. Each criterion maps to exactly one CLO for attainment reporting.

FYP-1 CLOs

CLO	Statement	Main Evidence
FYP1-CLO1	Defend a project idea by studying existing products, relevant literature/technical sources and current approaches, and by identifying a clear complex computing problem or non-trivial technical challenge.	Proposal, R&D brief, product/literature comparison, complexity justification
FYP1-CLO2	Define measurable requirements, scope, feasibility, risks and iteration plan for a two-semester project with clear member-wise responsibilities (demonstrating professionalism), DEI, ethical, and legal issues.	Requirements, acceptance criteria, risk register, iteration plan, DEI statement, ethical and legal statement
FYP1-CLO3	Design the proposed system using suitable architectural, data, process, interface and/or UML artefacts, and justify major design and technology choices.	Architecture, data/process artefacts, decision log
FYP1-CLO4	Implement an initial working prototype or core module and validate it using suitable tests, baseline checks or technical verification.	Live prototype, repository, tests, early validation
FYP1-CLO5	Identify and justify use of external tools, APIs, libraries, pretrained models or GenAI assistance and demonstrate human review, understanding and modification.	Tool disclosure, code/design walkthrough, Q&A
FYP1-CLO6	Present and defend the project effectively as a team while demonstrating traceable individual contribution.	Presentation, repo history, milestone log, supervisor feedback

FYP-2 CLOs

CLO	Statement	Main Evidence
FYP2-CLO1	Refine scope, requirements, design artefacts and implementation plan based on FYP-1 outcomes, supervisor feedback, panel feedback and controlled change requests.	Updated scope, revised design, change response
FYP2-CLO2	Implement and integrate modules into a coherent working system that addresses the identified complex computing problem or technical challenge.	Integrated system, live demo, repository, deployment evidence
FYP2-CLO3	Apply systematic testing and validation techniques to verify functionality, integration, performance, reliability, usability, security or model behavior, as applicable.	Test suite, validation logs, bug fixes, edge cases
FYP2-CLO4	Evaluate the developed system against baselines, existing approaches, user scenarios, datasets, benchmarks or clearly defined success criteria.	Evaluation results, comparison, failure/limitation analysis
FYP2-CLO5	Demonstrate responsible engineering practice in the use of external tools, APIs, libraries, pretrained models or GenAI assistance by explaining role, limitations, risks and human contribution.	Disclosure, dependency/risk register, technical defense
FYP2-CLO6	Demonstrate the final working system through effective presentation, live demo, technical defense and verifiable individual contribution across the lifecycle.	Final demo, report, repo history, individual ownership

2. Common Rules for All Forms

2.1 Scoring Scale

Score Band	Meaning
9–10	Strong evidence. Clear understanding, justified decisions, working evidence, validation and confident defense.
7–8	Good evidence. Minor issues exist, but ownership and reasoning are visible.
5–6	Basic evidence. Work exists, but explanation, validation or quality is limited.
3–4	Weak evidence. Superficial work, unclear understanding, poor traceability or incomplete demo.
0–2	No meaningful evidence, non-working item, hidden dependency, or student cannot explain the work.

2.2 Score Caps and Automatic Actions

Situation	Recommended Cap / Action
No live system at a demo stage where a live system is required	Implementation/ demo marks capped at 40% for that stage unless failure is justified with verified evidence.
Core feature is mainly a ready-made API/ tool/ template and no meaningful work is added	Core implementation and contribution marks capped at 50%; panel may require re-scope.
Undisclosed core use of GenAI/ API/ pretrained model/template is found	Responsible tool-use item = 0; refer to academic integrity policy if false declaration is confirmed.
Student cannot explain claimed owned module/ code/ design	Apply individual contribution adjustment. Group score may remain, but that member score should reduce.
Only polished report/ diagrams shown without design reasoning	Design/ artefact marks capped at 50%; ask for rationale and design evolution evidence.
Git commits exist but student cannot explain them	Do not give contribution credit from commit count alone. Use defense and walkthrough.

2.3 Responsible External Tool / API / GenAI Use Rubric

Level	Description	Assessment Treatment
L1: Supportive use	Tool used for grammar, formatting, brainstorming, debugging hints, documentation polish or minor code help.	Acceptable if not falsely presented as original technical contribution.
L2: Assisted development	Tool generated parts of code, tests, diagrams or text, but students reviewed, modified, integrated and can defend them.	Acceptable. Full marks possible if ownership, testing and modification ability are clear.
L3: Heavy core assistance	Tool/ API/ model used in an important module, but students add meaningful pipeline, customization, evaluation, error handling or integration.	Conditional. Full credit only with strong disclosure and evidence beyond tool use.
L4: Substituted work	Tool/ API/ template solves the core project, use is hidden, or students cannot explain/modify outputs.	Not acceptable. Apply caps, re-defense, re-scope or academic integrity process.

Form 1 — FYP-1 Proposal Defense Evaluation

Use: Week 4 proposal defense. Require POC-lite/ technical spike, not full MVP.
Recommended timing: *18–20 minutes per group maximum.*

Project Title: _____	Project ID: _____	Panel #: _____	Supervisor: _____	Role: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
Complex computing challenge and R and D basis	**FYP1-CLO1**	15	Problem is clear; gap is meaningful; complexity is not course-project level.	____	If No: do not approve without major revision or re-scope.
Product/ prior-FYP comparison and contribution clarity	**FYP1-CLO1**	5	Closest systems/ FYPs compared; difference is not cosmetic; contribution statement is explicit.	____	If weak: approve only with major modification or require re-defense.
Scope, requirements and success criteria	**FYP1-CLO2**	10	Major features/ modules and measurable acceptance criteria defined for two semesters and 2–3 members.	____	If vague: require scope freeze document before mid.
Proposed solution and technical approach	**FYP1-CLO3**	15	Architecture/ approach, algorithms/ models/ APIs/ data/ design choices are technically plausible and justified.	____	If tool solves core: cap and require re-scope.
POC-lite / technical spike / feasibility evidence	**FYP1-CLO4**	15	Clickable high-fidelity functional prototype, API/ data check, baseline run, sample pipeline or risk experiment shown. Full MVP not required.	____	If No: major modification or re-defense.
Responsible tool / API / GenAI plan	**FYP1-CLO5**	10	External tools/ APIs/ models/ GenAI disclosed; core/ support role and limitations understood.	____	If hidden/ unclear: require disclosure sheet before approval.
Work division and iteration plan	**FYP1-CLO2**	5	Member-wise responsibilities, milestones and first-semester deliverables are balanced; not only front-end work.	____	If weak: require revised iteration plan.
DEI, Ethical/ legal issues	**FYP1-CLO2**	5	DEI statement; Ethical/ Legal statement	____	If weak: require revised statements.
Presentation (Slides Quality, Delivery, Q&A)	**FYP1-CLO6**	20	Presentation design and delivery	____	If weak: low marks.

Decision	**Final comments / required improvement / individual concern**
Approved | Approved with minor modifications | Approved with major modifications, no re-defense | Re-defense required | Rejected / change idea	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 2 — FYP-1 Mid (Supervisory) Presentation and Demo Evaluation

Use: Around Week 8 of FYP-1. Focus on response to proposal feedback, early implementation, design reasoning and process evidence.

Recommended timing: *15 minutes per group.*

Project Title: _____	Project ID: _____	Panel #: _____	Supervisor: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
Previous proposal actions addressed	**FYP1-CLO6**	10	Panel action register shows each required change, status and evidence link.	____	If No: cap progress/ process marks and issue mandatory action.
Iteration plan and work breakdown progress	**FYP1-CLO2**	10	Planned milestone completed or delay is justified with recovery plan.	____	If weak: require revised plan with owners.
Live prototype with core module demo	**FYP1-CLO4**	25	Meaningful module works live; not only screenshots/ video. Edge/ failure case shown if possible.	____	If No: cap implementation/ demo marks.
Requirements and acceptance criteria quality	**FYP1-CLO2**	10	User interactions, FRs/ NFRs and acceptance checks are technically correct and aligned with scope.	____	If weak: cap SRS component and require correction.
Design artefacts and design evolution	**FYP1-CLO3**	15	Architecture plus data/ process/ behavior artefacts connected to implementation; changes since proposal are justified.	____	If diagrams only: cap design marks.
Repository (with CI/CD), milestone and member evidence	**FYP1-CLO6**	10	Tagged code, commit timeline, task log and member ownership visible. Commit count alone is not enough.	____	If No: cap process evidence at 40%.
Testing and early validation	**FYP1-CLO4**	10	Tests/validation checks exist for current module; one failure/ edge case discussed.	____	If No: mark testing low even if demo works.
Team dynamics and tool ownership	**FYP1-CLO5**	10	Team coordination and tool-assisted outputs are defended.	____	If one member weak: use contribution adjustment.

Decision	**Final comments / required improvement / individual concern**
Continue as planned | Continue with required corrections	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 3 — FYP-1 Mid (Supervisory) Report Evaluation

Use: Supervisor review of the FYP-1 mid report.
Recommended timing: *Report review outside presentation time.*

Project Title: _____	Project ID: _____	Panel #: _____	Supervisor: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
Report format and required artefacts present	**FYP1-CLO2**	5	Prescribed format followed; required sections and artefacts included. Compliance gate — cap relevant sections if No.	____	If No: cap relevant report sections.
Problem formulation, R&D and existing solution analysis	**FYP1-CLO1**	15	Report shows what was studied, what gap remains and what complex challenge is addressed.	____	If weak: require R and D rewrite.
Project vision, scope, modules and success criteria	**FYP1-CLO2**	10	Scope, modules, success criteria and feasibility are clear and consistent with approval.	____	If vague: require scope table.
Requirements and acceptance criteria	**FYP1-CLO2**	15	User interactions, FRs/ NFRs and acceptance checks are precise and testable.	____	If weak: cap SRS marks.
Architecture and component/ interface design	**FYP1-CLO3**	15	Architecture, components, dependencies and interfaces are clear and justified.	____	If only generated diagram: cap design marks.
Data/ process/ behavior design	**FYP1-CLO3**	10	Appropriate ERD/ schema/ data flow/ activity/ sequence/ state diagram included and explained.	____	If irrelevant: require project-specific artefact.
Implementation documentation of completed module	**FYP1-CLO4**	15	Completed module described with setup, main files/ classes/ functions and technical decisions.	____	If not reproducible: cap implementation documentation.
Testing/ validation plan and tool/ risk disclosure	**FYP1-CLO5**	15	Test plan, early results, external dependency disclosure, risk/ security/ privacy and limitations documented.	____	If disclosure missing: responsible tool-use item = 0 until submitted.

Decision	**Final comments / required improvement / individual concern**
Accepted | Accepted with corrections | Major correction required before next stage	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 4 — FYP-1 Final Presentation and Demo Evaluation

Use: End of FYP-1. Focus on implemented work, testing and readiness for integrated FYP-2 work.
Recommended timing: *15 minutes per group.*

Project Title: _____	Project ID: _____	Panel #: _____	Supervisor: _____	Role: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
Poster	**FYP1-CLO6**	10	Poster format, layout, style, contents	____	If weak: score should be low.
Previous mid/proposal actions addressed	**FYP1-CLO6**	10	Required changes from panel/ supervisor are addressed or formally justified.	____	If No: cap progress and issue carry-forward action.
Implementation progress against approved scope	**FYP1-CLO4**	25	At least 60% of committed work meaningfully implemented unless scope formally revised.	____	If No: cap implementation and consider re-demo.
Quality and robustness of live demo	**FYP1-CLO4**	15	Completed modules work beyond happy path; workflow stable enough for review.	____	If demo-only/ hardcoded: cap demo quality.
Testing on completed modules	**FYP1-CLO4**	10	Functional, edge, integration or model tests are executed and explained.	____	If No: testing score should be low.
Design evolution and updated artefacts	**FYP1-CLO3**	10	Major design/ requirement changes since mid are justified and reflected in artefacts.	____	If outdated artefacts: cap design marks.
Code quality, modularity and integration readiness	**FYP1-CLO4**	10	Code is structured, readable, buildable and ready for FYP-2 integration.	____	If weak: require integration-readiness action.
Responsible tool / API / GenAI ownership defense	**FYP1-CLO5**	5	Students explain external assistance, generated parts, validation and human modifications.	____	If weak: cap technical contribution.
Presentation and Q and A, team coordination	**FYP1-CLO6**	5	Short clear presentation, balanced participation, good handling of Q and A.	____	If member weak: use contribution adjustment.

Decision	Final comments / required improvement / individual concern
Continue as planned | Continue with required FYP-2 corrections	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 5 — FYP-1 Final Report Evaluation

Use: End-of-FYP-1 report. Must match demonstrated work and FYP-2 plan.
Recommended timing: *Report review outside presentation time.*

Project Title: _____	Project ID: _____	Supervisor: _____	Role: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
Report matches demonstrated system	**FYP1-CLO4**	5	Report reflects actual implemented modules, not only intended scope. Gate check: cap implementation documentation if No.	____	If No: cap implementation documentation.
Updated requirements and traceability	**FYP1-CLO2**	15	Requirements map to implemented features, pending features and acceptance criteria.	____	If weak: require traceability table.
Updated design artefacts with rationale	**FYP1-CLO3**	20	Architecture/ data/ process/ behavior artefacts reflect actual system and design decisions.	____	If outdated: cap design marks.
Implementation details and reproducibility	**FYP1-CLO4**	20	Setup, dependencies, configuration, main modules and technical flow clear enough to reproduce.	____	If not reproducible: cap implementation docs.
Testing evidence and early validation	**FYP1-CLO4**	20	Test cases include inputs, expected/ actual results, bugs/ fixes and edge cases.	____	If No: testing marks low even if demo works.
R&D continuation and FYP-2 evaluation plan	**FYP1-CLO1**	10	Evaluation plan includes baseline, user testing, benchmark, metric or comparison in FYP-2.	____	If weak: require evaluation plan before FYP-2 mid.
Tool / API / GenAI disclosure, risks and limitations	**FYP1-CLO5**	10	External assistance documented with core/support role and validation method.	____	If missing: responsible tool-use item = 0 until submitted.

Decision	**Final comments / required improvement / individual concern**
Accepted | Corrections required | Major correction required before FYP-2 mid	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 6 - FYP-1 Startup Evaluation Form (Optional for FYPs)

Evaluator Name & Signature: _____
FYP Name: _____
Date: _____
Total Score: _____

Category	Evaluation Criteria	Score (1–10)
Problem, Ideas and Need	Has the problem been identified? Disruptive and sustainable idea? How big is the pain? Short term or sustainable?	
Target Market	Clearly defined target market? Market size? Stable market or high growth potential?	
Solution	Does the solution address the problem? Is it scalable? Better/faster/cheaper? Clear value proposition?	
Team, Linkages and Advisors	Industry knowledge, unique skills, leadership experience, key relationships	
Competition	Barriers to entry, differentiation, future proofing, patents/proprietary solutions, key risks	
Go-To-Market Plan	Market and product identified? Market research completed? Go-to-market plan? Distribution channels?	
Business Model and Growth	Commercial and technical viability, growth and monetization strategy, MVP, LTV, CAC, churn, cash flow, traction	

Industry / Tech (circle/check applicable):

☐ Agri Tech
☐ AI / Machine Learning
☐ Big Data and Analytics
☐ Blockchain
☐ E-commerce
☐ Ed-Tech
☐ Fashion and Apparel
☐ Food-Tech
☐ Fin-Tech
☐ Gaming / AR / VR
☐ Healthcare / Medical
☐ IoT / Hardware
☐ Media / Content / Social Networks
☐ OTT and Communications
☐ Productivity
☐ Real Estate and Construction
☐ Robotics / Wearables / Implantables
☐ Shared / Gig Economy
☐ Social Enterprises
☐ Tourism and Travelling
☐ Other

Comments: _____

Form 7 — FYP-2 Mid / Pre-Job Fair Evaluation

Use: Around Week 8 of FYP-2. Focus on integrated system readiness and job-fair suitability.
Recommended timing: *12–15 minutes per group.*

Project Title: _____	Project ID: _____	Panel #: _____	Supervisor: _____	Role: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
FYP-1 final actions addressed	**FYP2-CLO1**	10	All carried actions are resolved or justified with evidence.	____	If No: revision/re-defense may be required.
Integrated live system is available	**FYP2-CLO2**	25	Major modules integrated and committed milestones mostly complete. No demo video only.	____	If No: normally revision/re-defense and cap demo marks.
Product workflow quality and robustness	**FYP2-CLO2**	15	System handles realistic scenarios and does not depend on hidden demo data only.	____	If fragile: recommend correction before job fair.
Integration quality across modules	**FYP2-CLO2**	15	Data/control flow across modules is coherent; dependencies and interfaces are managed.	____	If weak: cap integration marks.
Code quality, maintainability and deployment/ build evidence	**FYP2-CLO2**	15	Code is modular, readable, reproducible and deployable/buildable with documented setup.	____	If weak: issue code quality action.
Testing/ evaluation evidence and unresolved risk handling	**FYP2-CLO3**	10	Integration tests, validation results, known bugs and risk mitigation shown.	____	If missing: not ready for confident job fair recommendation.
Individual ownership and responsible tool-use evidence	**FYP2-CLO5**	10	Members explain owned modules; external APIs/ GenAI/ pretrained models disclosed and defended.	____	If weak: use contribution adjustment.

Decision	**Final comments / required improvement / individual concern**
Recommended for Job Fair | Recommended with minor corrections | Revision needed / re-defense	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 8 — FYP-2 Final (Supervisory) Presentation and Demo Evaluation

Use: Final evaluation. Focus on final working system, evaluation results and ownership.
Recommended timing: *10–12 minutes per group.*

Project Title: _____	Project ID: _____	Supervisor: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
FYP-2 mid actions addressed	**FYP2-CLO1**	5	Last required actions were addressed or justified.	____	If No: consider revision/re-demo depending on severity.
Final system functionality and completeness	**FYP2-CLO2**	25	System satisfies approved scope and handles key user/ system workflows.	____	If incomplete: cap final functionality.
Integration, reliability and deployment quality	**FYP2-CLO2**	15	Modules integrated coherently; system stable, reproducible and not demo-only.	____	If weak: cap integration/reliability.
Technical contribution and complex computing challenge solved	**FYP2-CLO4**	15	Final work addresses the non-trivial computing challenge, not only UI/ CRUD/ integration. Evaluated against defined success criteria.	____	If weak: cap technical contribution.
Testing, validation and evaluation results	**FYP2-CLO3**	20	Evaluated against tests, baselines, user scenarios, datasets, benchmarks or success criteria.	____	If No: evaluation marks low even if demo works.
Code quality, security, maintainability and reproducibility	**FYP2-CLO5**	10	Clean structure, error handling, basic security/ privacy, setup/ deployment documentation.	____	If weak: deduct from quality.
Individual technical defense and tool-use accountability	**FYP2-CLO6**	10	Each member explains owned work and external tool/ API/ GenAI role with limitations.	____	If member weak: use contribution adjustment.

Decision	**Final comments / required improvement / individual concern**
Pass | Pass with comments | Revision/re-demo required | Best FYP nomination: Yes / No / Maybe	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Form 9 — FYP-2 Final Report Evaluation

Use: Final report assessment. Focus on final traceability, implementation, testing, evaluation and limitations.
Recommended timing: *Report review outside presentation time.*

Project Title: _____	Project ID: _____	Supervisor: _____	Role: _____	Date: _____

Integrated Gate-Check, Scoring and Action Table

Criterion / Gate Item	CLO	Max	What to Ask or Observe	Score	Action if No / Weak Evidence
Report matches final system	**FYP2-CLO6**	5	Report matches actual delivered system, not only planned scope. Verifiable individual contribution across the lifecycle.	____	If No: cap implementation/evaluation documentation.
Final problem, R&D context, scope and contribution	**FYP2-CLO1**	10	Problem, product/ research basis, gap, contribution and limitations are clearly stated.	____	If weak: cap R&D/contribution marks.
Requirements, design and implementation traceability	**FYP2-CLO2**	15	Requirements connect to design artefacts, code modules and tests.	____	If weak: require traceability correction.
Implementation architecture and reproducibility	**FYP2-CLO2**	20	Setup, dependencies, APIs, data/ model pipeline and key modules are clear.	____	If not reproducible: cap implementation docs.
Testing and validation results	**FYP2-CLO3**	20	Functional, integration, edge, performance/ security/ usability/ model tests documented with outcomes.	____	If missing: testing marks low.
Evaluation against baseline/success criteria	**FYP2-CLO4**	20	Evidence shows usefulness against baseline, existing approach, user need or benchmark.	____	If No: evaluation marks low.
Responsible tool / API / GenAI use, ethics, limitations and future work	**FYP2-CLO5**	10	External assistance, limitations, privacy/ security/ ethics and future work honestly documented.	____	If missing: responsible use marks = 0 until corrected.

Decision	**Final comments / required improvement / individual concern**
Accepted | Accepted with minor corrections | Major report correction required	Deduction reason: _____
	Required improvement / next action: _____
	Individual concern, if any: Member(s) ____ Evidence ____

Auxiliary Forms (10–13)

Forms 10 (Panel Action Register), 11 (Individual Contribution Adjustment), 12 (Controlled Change Request), and 13 (External Tool / API / GenAI Disclosure Sheet) are supporting instruments used alongside the main evaluation forms.

Form 10 - Panel Action Register

Use this as the carried record of panel/supervisor feedback. Students update it before every evaluation. Panel only checks status and evidence link, not long explanations.

Action ID	Raised at	Required action/ criticism	Owner(s)	Due by	Status	Evidence link/ note	Panel status
A1					Open / Done / Partly / Rejected		Accepted / Carry forward
A2					Open / Done / Partly / Rejected		Accepted / Carry forward
A3					Open / Done / Partly / Rejected		Accepted / Carry forward
A4					Open / Done / Partly / Rejected		Accepted / Carry forward

Rule: later evaluation forms should not repeat every old comment. They should score whether this register was addressed with evidence.

Form 11 - Individual Contribution Adjustment Form

Use only when there is evidence of unequal contribution, free-riding, inability to explain owned work, or false ownership. Do not use commit count alone.

Member	Owned module/artefact claimed	Evidence checked	Defense result	Multiplier
Member 1		Repo / demo / report / Q&A	Strong / Adequate / Weak / Failed	1.00 / 0.85 / 0.70 / 0.50 / 0.00
Member 2		Repo / demo / report / Q&A	Strong / Adequate / Weak / Failed	1.00 / 0.85 / 0.70 / 0.50 / 0.00
Member 3		Repo / demo / report / Q&A	Strong / Adequate / Weak / Failed	1.00 / 0.85 / 0.70 / 0.50 / 0.00

Multiplier	Meaning
1.00	Clear meaningful contribution and can defend owned work.
0.85	Minor gaps in ownership or contribution evidence, but acceptable participation.
0.70	Weak or uneven contribution; limited explanation of owned work.
0.50	Major contribution concern; student cannot explain important claimed work.
0.00	No meaningful contribution, absent from defense without approval, or false ownership confirmed.

Suggested calculation: individual stage score = group stage score x multiplier. Supervisor and panel may add a short justification where needed.

Form 12 - Controlled Change Request

Use this to keep scope flexible without allowing unlimited faculty wishes. One bounded change may be issued at FYP-1 mid, FYP-1 final and FYP-2 mid. No new scope at FYP-2 final.

Field	To be completed by panel/supervisor
Change request	_____
Reason for request	Design weakness / missing edge case / quality issue / validation gap / API risk / other: _____
Expected evidence by next evaluation	Code change / test case / design update / report update / demo scenario / evaluation result
Difficulty and time expectation	Small / Medium / Large. Estimated effort: _____ days. Should not change full project direction.
Owner(s)	Member(s): _____

Acceptance test	How panel will know it is done: _____

A controlled change should test understanding and quality. It should not add a new product line, new platform, or unrelated feature late in the project.

Form 13 - External Tool / API / GenAI Disclosure Sheet

Student-submitted sheet. It should be updated before every evaluation. It is not a prompt journal; it is a contribution boundary document.

Tool/ API/ model/ template used	Used in which part?	Support/core category	Student contribution beyond tool	Validation / risk control
		Support / Core-assist / Core-replacement		
		Support / Core-assist / Core-replacement		
		Support / Core-assist / Core-replacement		
		Support / Core-assist / Core-replacement		
		Support / Core-assist / Core-replacement		

Student declaration: We confirm that the above disclosure is accurate to the best of our knowledge. Hiding core tool/ API/ GenAI use may lead to mark deduction, re-defense, re-scoping or academic integrity action.

Member	Signature	Date
Member 1		
Member 2		
Member 3		

Sample Evaluator Question Bank

Panel members may pick 3-5 questions per group from the following list (but not limited to these only).

Area	Some sample questions
Problem and R&D	What existing solution is closest? What gap remains? Which recent work or product influenced your design? What baseline will you compare against?
Complex computing challenge	Which part is technically hard? Why is it not a course project? Which constraints make this non-trivial?
Requirements	Pick one major requirement and show its acceptance test. What changed since proposal and why?
Architecture	Why this architecture? What breaks if this component fails? Which dependency is most risky?
Code ownership	Show the file/ function that implements this feature. Who wrote/reviewed it? What would change if the rule changes?
Tool/ API/ GenAI	What exactly did the tool/ API/ model do? What did you add beyond it? How did you verify output? What is the fallback?
Testing/ evaluation	Show one failed test or edge case. What metric proves usefulness? What limitation remains?
Individual contribution	Ask each member to explain one owned module and one design decision they personally made.

Industry-collaborative FYP rule

- Industry deliverables may be accepted as milestones only if mapped to academic evidence: problem analysis, design reasoning, implementation, testing, evaluation and individual contribution.
- If source code cannot be stored in the university GitHub due to NDA/IP concerns, supervisor and FYP committee must still receive private evidence or repository access.
- Client satisfaction alone should not replace academic assessment. Students must defend technical decisions, testing, limitations and contribution boundaries.
- Proprietary data, API keys or private client assets must not be committed to public/ shared repositories.

CLO-to-Form Mapping Summary

Each cell shows which criteria in that form map to the named CLO. Every criterion carries exactly one CLO label.

Form	FYP1-CLO1	FYP1-CLO2	FYP1-CLO3	FYP1-CLO4	FYP1-CLO5	FYP1-CLO6
Form 1 Proposal	CCP & R and D idea/ product comparison	Scope and requirements Work division & plan	Technical approach	POC-lite/ spike	Tool/ GenAI plan	—
Form 2 FYP-1 Mid Pres.	—	Iteration plan Requirements quality	Design artefacts	Live prototype Testing and validation	Presentation and tool ownership	Previous actions Repo and member evidence
Form 3 FYP-1 Mid Report	Problem and R and D	Report format Scope and requirements	Architecture Data/ process design	Implementation docs	Testing plan and tool disclosure	—
Form 4 FYP-1 Final Pres.	—	—	Design evolution	Implementation progress Demo quality Testing Code quality	Tool ownership defense	Previous actions Communication
Form 5 FYP-1 Final Report	R and D continuation and evaluation Plan	Requirements and traceability	Updated design artefacts	Report matches system Implementation details Testing evidence	Tool disclosure	—

Form	FYP2-CLO1	FYP2-CLO2	FYP2-CLO3	FYP2-CLO4	FYP2-CLO5	FYP2-CLO6
Form 7 FYP-2 Mid	FYP-1 actions addressed	Integrated system Workflow quality Integration Code quality	Testing and risk evidence	—	Ownership and tool-use evidence	—
Form 8 FYP-2 Final Pres.	FYP-2 mid actions	Final functionality Integration and deployment	Testing and evaluation results	Technical contribution and CCP solved	Code quality and security	Individual defense and tool accountability
Form 9 FYP-2 Final Report	R&D context and contribution	Traceability Implementation and reproducibility	Testing and validation results	Evaluation vs baseline	Tool/ GenAI ethics and future work	Report matches final system

Appendix C: FYP Idea Selection and Proposal Readiness Guide

Core standard

Every FYP, regardless of stream, domain or supervisor, must demonstrate six connected elements: (1) research-informed problem understanding; (2) a clearly justified complex computing problem or complex computing activity; (3) a coherent designed and implemented system that runs in a realistic deployment environment; (4) systematic testing and evaluation; (5) the engineering documentation set that forms the project report; and (6) traceable team and individual ownership. The idea should also state a relevant SDG or thematic alignment as broader context. This alignment is not a substitute for CCP, contribution, feasibility or evaluation. An idea may enter through published research or through an observed real-world problem, but both points are judged against this same standard. These are not separate tracks; they form one common academic mechanism for all FYPs.

Important policy note: Permission to use any external tool, service, model, API, dataset, library, template, or third-party component is governed by current university policy on academic integrity, ethics, plagiarism, disclosure, licensing, copyright, privacy and security. This guide neither grants nor withholds that permission. What this guide defines is the academic standard that applies to whatever use is permitted.

External components are expected, not exceptional: Modern computing practice is built on frameworks, libraries, services, pre-trained models, hosted APIs and templates, and their use in a final year project is normal and expected. The boundary is not whether an external component is used, but what it supplies. An external component may support the project. It may not supply the complex computing problem the group claims to have solved. Section 14 sets out how that boundary is classified and judged.

Regardless of how any part of the work was produced, students remain individually responsible for the reasoning, design, implementation, verification and defense of what they submit. Authorship of keystrokes is not the test a panel applies; understanding, justification and the ability to modify the work under questioning are. Permitted use that is properly disclosed and acknowledged is not penalized. Undisclosed use, or work that cannot be explained by the member who claims it, is treated as an integrity or ownership failure.

How to use this guide:

Sections 1 to 17 apply to every FYP without exception and should be read first. Annex A adds guidance for groups whose idea begins from published research. Annex B adds guidance for groups whose idea begins from an observed real-world or everyday problem. Annex C shows worked examples of complex computing problems formulated at three levels of maturity, and applies to both streams. The annexes assume the common sections and do not repeat them.

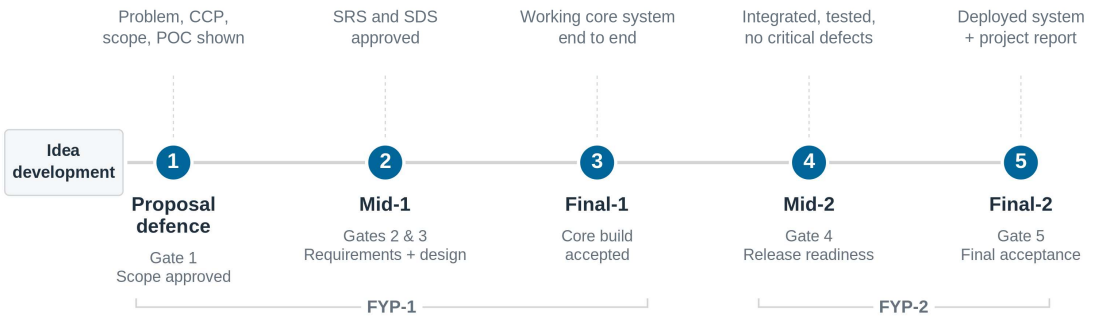


Figure 1. The FYP at a glance — from idea development through five evaluation stages to a deployed system and project report.

C.1. What kind of FYP should you select?

A Final Year Project (FYP) is the capstone of computing degrees. It should not be selected merely because an application can be built around it. A suitable idea begins with a meaningful problem, develops a defensible technical contribution, and provides enough evidence for the panel to judge both the quality of the product and the learning of the students.

- Every FYP must be research-informed and evaluated, not only implemented.
- Every FYP must produce a working, reproducible computing artefact that can be deployed and operated outside the development machine and not only a literature review, a notebook of results or a recorded demonstration.
- Every FYP must produce the engineering documentation set that culminates in the project report, and the idea must be specific enough to be written down in that form.
- An industry-collaborative FYP must still satisfy every academic requirement, even when the client mainly values delivery.
- A data, AI, security, networking, HCI, IoT, software engineering or infrastructure FYP is judged by the same core mechanism, although the evidence will differ by domain.

C.1.1 Two Entry Points, One Standard

The School recognizes two entry points into an FYP idea. Stream A begins from published research. Stream B begins from an observed real-world or everyday problem. The stream describes where the idea came from and not how difficult it is, not how prestigious it is, and not how it will be graded. Both streams converge on the same obligations: a defensible complex computing problem, a working deployable system, systematic evaluation, a complete documentation set and traceable individual ownership. A group should declare its stream at proposal stage so that the panel can ask the right questions and may change it later only with supervisor approval.

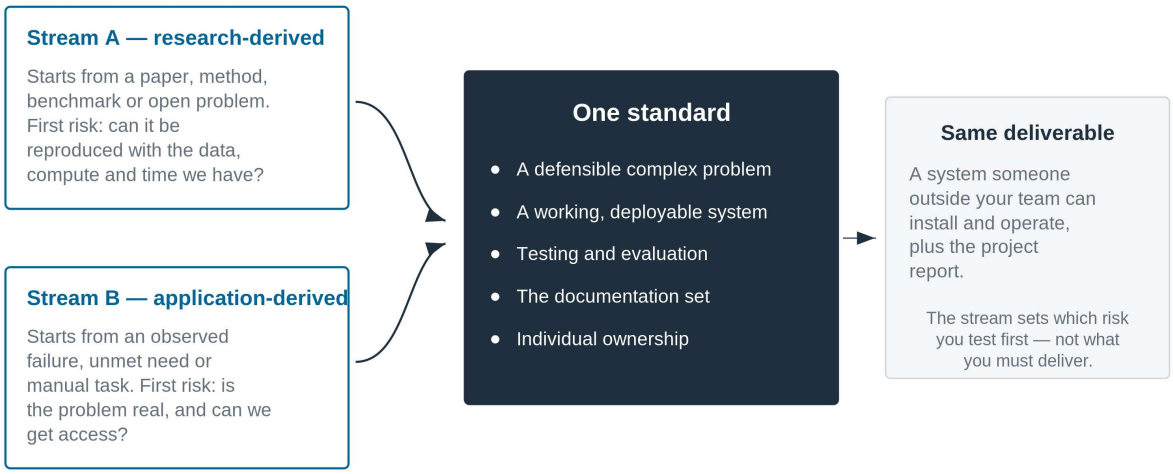


Figure 2. Two entry points, one standard. The stream determines which risk you test first, not what you must deliver.

Aspect	Stream A — research-derived	Stream B — application-derived
Starting point	One or more peer-reviewed papers, a published method, a benchmark, a dataset or an open problem stated in the literature.	An observed workflow failure, an unmet stakeholder need, or an everyday task that is currently handled manually or badly.

Aspect	Stream A — research-derived	Stream B — application-derived
Where the gap comes from	A stated limitation, an unexamined assumption, an untested setting, a missing comparison or an unaddressed constraint in the published work.	Evidence from stakeholders, observed practice, product documentation, standards and prior FYPs showing what remains unsolved or inadequately solved.
Primary risk to test early	Whether the method can be reproduced, adapted or even run within the data, compute, license and time budget actually available to the group.	Whether the problem is real, whether stakeholders will engage, and whether the required data, access, permissions or integration can be obtained.
Typical contribution	Transparent reproduction, adaptation to a new context or constraint, comparison of methods, failure analysis, or engineering a published method into an operable system.	Problem or requirement formulation, system and architecture design under real constraints, workflow or accessibility improvement, or a curated domain resource.
Most common failure	Re-running someone else's code without understanding it, and finishing with results but no system that anyone could operate.	Building a competent application around a problem that never required computing depth, and discovering at defense that no CCP can be identified.
What the panel will press on	Can you explain, modify and defend the method without the code in front of you? What is your delta over the paper? Does it become a real system?	Where exactly is the complexity? What would a routine implementation have got wrong? What evidence shows this problem is real?

Both streams end in the same place

A Stream A project that produces excellent results, but no operable system has not met the standard. A Stream B project that produces a polished application with no research basis, no defensible complex computing problem and no evaluation beyond a demonstration has not met the standard either. The stream determines where a group starts looking and which risk it tests first. It does not reduce what the group must deliver.

C.1.2 Working, Deployable System

An FYP is not complete when the code runs on the developer's laptop. The final artefact must be installable and operable by someone other than its authors, in an environment that reasonably represents its intended use. This is a requirement of engineering competence rather than of commercial success: the panel assesses whether the system is deployable, documented and defensible, not whether it has users or revenue. Groups should nevertheless be able to say who would use the system, what adoption would require and what stands in the way. That thinking almost always improves the requirements, and it frequently exposes the constraints where the real computing complexity lives.

Deployment readiness dimension	What should be true by FYP-2
Runs outside the development machine	The system can be installed and started from documented steps, on a clean environment, by someone who did not build it.

Deployment readiness dimension	What should be true by FYP-2
Realistic operating environment	The deployment target is stated and justified e.g., cloud, on-premises, container, edge device, mobile, offline or embedded etc., and the system is deployed there rather than only demonstrated locally. Deployment to the real target is expected wherever it is technically and legally feasible; where it is not, the group deploys to the closest defensible substitute and states what remains untested.
Configuration and data	Configuration, credentials, secrets and seed or sample data are handled deliberately rather than hard-coded, so that the system starts from a known, reproducible state.
Non-functional behavior	Performance, capacity, reliability, security, privacy and accessibility behavior appropriate to the problem is stated as measurable criteria and tested, not asserted.
Failure and recovery	Known failure modes, degraded behavior, and recovery or fallback paths are identified and demonstrated, including what happens when an external dependency is unavailable.
Handover	Source, build and run instructions, dependencies, licenses, known limitations and future work are packaged so that another party could continue, evaluate or operate the project.
Adoption context	Intended users, the realistic path to use, and the principal barriers such as cost, data access, integration, regulation, trust or maintenance are identified honestly. This is assessed as engineering judgement, not as commercial achievement.
Evidence of deployment	A record of the deployed system: where it runs, how it was installed, who can access it, and what it cost to stand up. A screenshot of a local build is not deployment evidence.
Deployability is required; commercial success is not **Groups are not expected to launch a product, acquire users or demonstrate revenue. They are expected to build something that could credibly be handed to a real user or operator. Where a project genuinely cannot be deployed into its intended environment — for example where clinical, legal, safety or regulatory approval would be required. The group must deploy into the closest defensible substitute and state explicitly what remains untested, and why.**	

C.1.3 SDG or Thematic Alignment

Every group should identify one relevant UN Sustainable Development Goal (SDG) or one relevant thematic area such as health, education, environment, smart cities, innovation, accessibility, safety, public services, industry or community welfare. The link should be explained in two or three sentences by connecting the project problem, stakeholders, and expected contribution to that theme.

Do not use SDG or thematic alignment to decorate a weak idea. A broad label such as “education” or “innovation” does not prove that the project is complex, useful or academically strong. The alignment only gives wider context; the project must still defend its problem, CCP or complex activity, contribution, POC, evaluation and ownership.

Field	What students should write
Primary SDG or thematic area	Select one main SDG or theme that naturally matches the project problem.
Connection to problem	Explain which stakeholder problem or consequence creates this alignment.
Boundary	State clearly that the SDG/ theme is context and not the claimed technical contribution or CCP.

C.2. Research and Development is Embedded in Every FYP

Research in an undergraduate FYP does not always mean inventing a new algorithm or publishing a paper. It means that the project is informed by credible evidence rather than assumptions. The group must study what already exists, understand relevant technical knowledge, use that understanding to make decisions, and define how the proposed solution will be evaluated.

R and D element	What it means in a student FYP	Evidence expected at proposal stage
Solution landscape	Study existing commercial products, open-source systems, prior FYPs, manual workflows and relevant standards.	A comparison that identifies what is already solved, what remains weak, and why the proposed work is not a copy.
Technical foundation	Study relevant methods, architecture, algorithms, datasets, protocols, standards, design principles or domain knowledge.	A focused review showing which sources influence the proposed approach and which alternatives were considered.
Reproduction, adaptation or experimentation	Where suitable, reproduce or adapt a published method, benchmark alternatives, or test a technical assumption.	A plan for implementation/ experimentation and a clear statement of what will be reproduced, changed or compared.
Evaluation and learning	Define evidence that will show whether the solution is correct, useful, reliable or better suited to the problem.	Baselines, test scenarios, metrics, user studies, benchmarks or acceptance criteria appropriate to the project.

State-of-the-art work is welcomed: two forms of engagement with recent, high-performing research are particularly encouraged. The first is implementing a state-of-the-art method as a deployable product. Published work very often stops at a reported result on curated data, and the engineering required to turn that method into something a user or operator can actually run — under real input, real latency, real cost, real failure and real deployment constraints — is substantial, defensible and frequently where the complex computing problem actually lives. The second is improving the methodology of a state-of-the-art paper at a gap the group has identified through its own study of the literature, and then demonstrating that improvement against the original method as a baseline. Both are strong bases for an FYP. Both must still end in a working, deployable system and a defensible evaluation.

What is not required in every FYP: novelty is not a compulsory claim. A group is not obliged to invent a new algorithm or to beat the best published number, and Section 4 sets out several other contribution types that are fully acceptable when they are substantial and properly justified. What is required at every level of

ambition is understanding. Implementing or improving a state-of-the-art method without being able to explain its mechanism, its assumptions and its failure modes is not acceptable, however impressive the result appears.

The selected sources should be current and authoritative for the topic, while seminal papers, established standards and official documentation may also be necessary. What matters is that the research changes or justifies the project's problem formulation, design or evaluation. A literature section that has no visible effect on the project is not sufficient.

C.3. Understanding a Complex Computing Problem (CCP)

The Seoul Accord defines a complex computing problem as a computing problem having some or all of a set of recognized characteristics. It does not require every characteristic, and it does not prescribe a fixed number that must be ticked. However, a credible CCP claim should be central to the project and normally be supported by several meaningful characteristics, not by one superficial statement.

Seoul Accord Characteristic	**Question for your idea**	**How to interpret it?**
1. Conflicting requirements	Are there important technical, computing, organizational, legal, ethical, economic or user requirements that pull the solution in different directions?	Example: accuracy versus latency, privacy versus personalization, security versus usability.
2. No obvious solution or depth of analysis	Is there no standard answer that can simply be copied? Does the group need conceptual thinking, modelling or comparative analysis?	A standard CRUD workflow normally does not satisfy this characteristic.
3. Depth of knowledge	Does the solution require in-depth computing or domain knowledge and reasoning based on established principles?	The group should identify the computing and domain knowledge it must learn and apply.
4. Unfamiliar issues	Does the project involve issues that are not frequently encountered in routine development?	Unfamiliarity should come from the problem, constraints or analysis, not from choosing an unfamiliar framework.
5. Beyond standard practice	Is the central problem outside what can be solved completely through standard procedures, templates or ordinary professional practice?	Using a standard framework is acceptable, but it should not solve the core challenge by itself.
6. Diverse stakeholders	Are there stakeholder groups with significantly different needs, authority, risks or success criteria?	The project must show how these needs affect requirements and design.
7. Significant consequences	Can poor performance, wrong decisions, insecurity or failure have important consequences?	The evaluation and safeguards should match the seriousness of the consequences.
8. Interdependence	Is the problem high-level, with multiple interacting sub-problems whose behavior affects the whole system?	A large number of independent screens is not the same as interdependent technical sub-problems.

Seoul Accord Characteristic	Question for your idea	How to interpret it?
9. Ill-defined requirements or cause	Is part of the requirement, root cause or correct solution initially unclear and in need of investigation?	Students should show how research, stakeholder study, experiments or a POC will reduce this uncertainty.

C.3.1 Range of Computing Activities

The Seoul Accord defines the following range of computing activities:

Seoul Accord Characteristic	A **Complex Computing Activity** is a computing activity or project that has some or all of the following characteristics:	A **Broadly-defined Computing Activity** is a computing activity or projects that has some or all of the following characteristics:	A **Well-defined Computing Activity** is a computing activity or project that has some or all of the following characteristics:
1. Range of resources (people, money, equipment, materials, information, and technologies)	Involves the use of diverse resources	Involves a variety of resources	Involves a limited range of resources
2. Level of interactions	Requires resolution of significant problems arising from interactions among wide-ranging or conflicting technical, computing, contextual, or other issues	Requires resolution of occasional interactions among technical, computing, contextual, and other issues, of which few are conflicting	Requires resolution of interactions between limited technical and computing issues, with little or no impact from broader issues
3. Innovation	Involves creative use of knowledge of computing or domain principles in novel ways	Involves the use of new resources, techniques, or computing processes in innovative ways	Involves the use of existing resources techniques, or computing processes in new ways
4. Consequences to society and the environment	Has significant consequences in a range of contexts	Has consequences that are most important locally, but may extend more widely	Has consequences that are locally important and not far-reaching
5. Familiarity	Can extend beyond previous experiences by applying principles-based approaches	Requires a knowledge of normal operating procedures and processes	Requires a knowledge of practical procedures and practices for widely applied operations and processes

Do not manufacture complexity

A project should not add unnecessary technologies, modules or difficult features merely to appear complex. The CCP should arise naturally from a real problem and should lead to meaningful analysis,

design trade-offs and evaluation. Artificial complexity increases risk but does not improve academic quality.

Annex C shows twelve domains formulated at three levels of maturity, together with the characteristics each formulation exhibits. For most groups the contrast between a weak and a strong formulation of the same problem is more instructive than the table above.

Does a CCP make a project more interesting?

A CCP does not automatically make an idea attractive. It makes the idea academically defensible when the complexity is connected to a worthwhile problem. A project becomes interesting when students must make real decisions, resolve trade-offs, test uncertain assumptions and produce evidence. The purpose of CCP is therefore not to make the project look difficult; it is to explain why the project deserves capstone-level study and cannot be reduced to routine implementation.

Complex computing activity is also relevant

The Seoul Accord separately describes a complex computing activity or project through characteristics such as diverse resources, significant interactions among technical and contextual issues, creative use of knowledge, significant consequences, and work that extends beyond previous experience using principles-based approaches. A project may therefore be defensible through one central CCP and the complex activity required to solve it. Students should not claim several unrelated CCPs only to inflate scope.

C.4. Contribution: What Value Will Your FYP Add?

Contribution means the meaningful value created by the project beyond repeating what already exists. Novelty may strengthen a project, but novelty should not be treated as a compulsory claim of invention. A rigorous reproduction, adaptation, integration under difficult constraints, evaluation, or workflow improvement may be a valid contribution when it is substantial and properly justified.

Possible contribution type	**Examples**	**What must be shown**
Problem or requirement contribution	A clearer formulation of a neglected sub-problem; resolving an ill-defined workflow or stakeholder conflict.	Evidence that the problem is real, not already adequately solved, and that the formulation changes the solution.
Method or algorithm contribution	Reproducing, adapting, combining or comparing relevant techniques for a new context; improving the methodology of a state-of-the-art method at an identified gap; or engineering a state-of-the-art method into a deployable system.	A justified baseline — normally the original method — transparent adaptation, and evidence of performance, cost or limitations under the group's own evaluation.
System or architecture contribution	A reliable, secure, offline, distributed, edge, scalable or interoperable solution under meaningful constraints.	Why is ordinary architecture insufficient; key trade-offs, interfaces, risks and validation.
Data or resource contribution	A curated dataset, annotation process, benchmark, ontology, simulator or domain-specific data pipeline.	Quality controls, ethical/legal handling, documentation and how the resource supports the problem.
Human/workflow contribution	Human-in-the-loop decision support, accessibility, explainability, safer workflow or improved coordination.	Stakeholder evidence and evaluation of the actual workflow, not only interface appearance.

Possible contribution type	Examples	What must be shown
Evaluation contribution	A defensible benchmark, failure analysis, user study, comparison method or validation protocol.	Why the evaluation is suitable and what conclusions it can and cannot support.

Contribution statement

Use a precise sentence: "Existing approaches perform X, but remain weak under Y condition. We will contribute Z through A, and evaluate it using B." Avoid claims such as "we will use AI", "we will add RAG", "we will make it local", "we will improve the UI", or "we will combine many features" unless the technical contribution and evaluation are clearly explained.

C.5. Scope for the Approved Group Across Two Semesters

An FYP should be one coherent project, not a collection of loosely connected individual mini-projects. Scope is not judged mainly by the number of screens, technologies, features or estimated coding hours. It is judged by the depth of the problem, the amount of reasoning and engineering required, the quality of the evidence, and whether every registered member can make a meaningful and accountable technical contribution.

Scope principle	Practical meaning
One shared central problem	All members should be able to explain the same problem, contribution, architecture and evaluation story.
Meaningful primary ownership	Each member should have a substantial technical responsibility that includes decisions, implementation and validation. A member should not be limited to slides, documentation, routine UI or manual testing.
Interdependent work	Responsibilities should connect through defined interfaces, shared data, integration or common evaluation. The project should not become separate individual submissions.
Shared engineering responsibilities	Architecture, integration, quality assurance, ethics/ security, repository discipline and final defense remain team responsibilities even when one member leads a part.
Balanced depth, not identical tasks	Contributions need not be equal in type. One member may lead a data/ model pipeline, another a core system or service, and another an interaction/ deployment/ evaluation subsystem, provided each contribution is technically meaningful and the workload is reviewed over time.
Evolving division of work	Initial ownership can change as the design evolves. The team should maintain a task/ milestone record and be prepared to justify changes.

A useful scope test: If one member were removed, would a meaningful technical responsibility and its evidence disappear? If the answer is no for a member, the proposed division is probably too shallow. This is a diagnostic question, not a rule that each member must build a separate module.

C.5.1 Checkpoints: The Five Evaluation Stages

An FYP is assessed at five points. The proposal defense is the first, followed by Mid-1 and Final-1 in the first semester, and Mid-2 and Final-2 in the second. Groups should test a candidate idea against this trajectory before committing to it. An idea that cannot show demonstrated feasibility by the proposal defense or cannot reach a working core system by Final-1, is too large, too uncertain or too dependent on resources the group does not yet have.

These five points correspond closely to the approval gates defined in the School's project documentation standard, and the table below names both. The technical column states what should exist in the project itself; the documentation column states the state of the written record. Detailed assessment criteria and weightings are defined by the FYP evaluation forms, and the content and format of each document by the documentation standard; where this table and those documents differ, they take precedence.

Stage and gate	**What should exist in the project**	**Documentation state**
1. Proposal defense Gate 1 — Scope approved	A defined problem, an evidenced gap, a justified CCP, a contribution statement and an evaluation plan. A demonstrated POC or technical spike (Full MVP is not required). Committed scope separated from optional extensions, with primary technical ownership allocated to each member.	Scope of work approved — see 5.2 on who approves it. Project proposal and initiation plan complete. First version of the timeline, milestones and deliverables, with a named approver and an acceptance basis for each deliverable. Initial RAID register opened for risks, assumptions, issues and dependencies.
2. Mid-1 Gate 2 — Requirements baseline Gate 3 — Design baseline	Requirements baselined with stakeholders. Architecture chosen and justified against the alternatives considered, with key design decisions and their rationale recorded. Data, access, licenses and ethical clearance are secured. Live prototype with a core module implemented.	Software requirements specification approved: requirements uniquely identified, testable and prioritized, with at least one measurable non-functional requirement for every quality attribute that matters. Software design specifications approved, covering components, data design, interfaces, security, deployment topology and failure handling. Traceability matrix opened. API and third-party dependency register started.
3. Final-1 Exit gate — Core build accepted	A working core system covering the primary scenario end to end, built against the approved design. Integration path proven rather than assumed. The comparison baseline measured. Deployment target selected and a first deployment attempted.	Traceability matrix populated for all committed requirements, showing requirement to design to test coverage. Test, deployment, rollback and handover plan drafted, including defect severity definitions and release criteria. Timeline revised against actual progress, with any scope change processed under 5.3.
4. Mid-2 Gate 4 — Release readiness	Committed scope implemented and integrated. System testing complete. Non-functional criteria measured against their stated targets rather than asserted. Evaluation running against the baseline. Failure and edge cases exercised deliberately.	System test report with results and a defect list by severity: no open Critical defects, and every open High defect carrying an approved disposition. Deployment and rollback plans approved and rehearsed rather than described.

Stage and gate	What should exist in the project	Documentation state
5. Final-2 Gate 5 — Final acceptance	A deployable, documented system installed and operated in the target environment — or a defensible substitute — by someone outside the team. Acceptance testing completed with the supervisor or client. Completed evaluation with results, limitations and failure analysis. Every member able to defend their technical ownership.	Complete project report: all documents at their approved version, acceptance matrix signed off, and a handover package containing source, build and run instructions, dependencies, licenses, credentials handling, known limitations and future work. Any remaining items are recorded in a signed punch list with owners.

Every document in the set carries a document control block recording version, date, author and status, and moves from draft, through review, to approved baseline. Groups should version their documents rather than overwrite them. The panel should be able to see how the requirements and the design evolved across the two semesters, and why.

This mapping is proposed for review

Two judgements in this table should be confirmed before the guide is issued. First, both the requirements baseline and the design baseline are placed at Mid-1, which follows the documentation standard's rule that design is baselined before implementation — but it also determines how much implementation FYP-1 demands. Second, the documentation column is a proposed sequencing of the School's document set and should be checked against the FYP evaluation forms. Groups should in all cases follow the stage requirements published for their cohort.

C.5.2 Who Acts as the Client

The School's documentation standard is written for a client-and-vendor relationship: the scope of work is issued by a client, and the development team responds to it. Final year projects rarely have that relationship, so the role must be assigned explicitly at proposal stage rather than left ambiguous.

Project situation	Who acts as client
Industry-collaborative project	The partner organization acts as client and approves the scope of work. The supervisor remains the academic approver, and the client's satisfaction does not by itself satisfy the academic requirements.
Supervisor-proposed project	The supervisor acts as client for scope approval while remaining the group's supervisor. The group should record which decisions were taken in which capacity.
Group-originated project (most Stream B ideas)	The group writes the scope of work on behalf of a named, real stakeholder it has actually consulted, and the supervisor approves it. An invented client is not acceptable.
Research-derived project (most Stream A ideas)	The intended operator or user of the resulting system acts as the notional client. The scope of work states who would run the system and what they would need from it.

In the commercial model the client supplies a prioritized list of functional and non-functional requirements and the vendor implements it. In an FYP, the group must discover the requirements itself — through stakeholder study, the solution landscape and its own analysis — because that discovery is a substantial part of the academic work being assessed. A group that is handed a requirements list has been handed the part of the project that carries the marks. Where an industry partner does supply requirements, the group must still establish the problem, the gap, the CCP and the evaluation independently, and should state plainly which requirements were given and which it derived.

C.5.3 Change Control after Approval

Approval at the proposal defense baselines the scope. Projects change after that — a design proves unworkable, data turns out to be unavailable, an assumption fails — and changing course in response to evidence is a sign of good engineering rather than of failure. What is not acceptable is silent change: quietly dropping a committed requirement, substituting an easier problem, or expanding scope without assessing the consequence.

Once a document reaches its approved version, any change affecting scope, schedule, architecture, security, data handling or acceptance criteria should be recorded and approved by the supervisor before it is acted upon. The record need not be elaborate — what changed, why, what evidence prompted it, what it affected and who approved it — but it must exist, and the panel should be able to trace the final system back to the approved proposal through it.

Change requiring a recorded decision	**Why it matters**
Removing or deferring a committed requirement	The committed scope is what the project was approved against. Deferral is acceptable when justified and recorded; silent removal is not.
Changing the core method, architecture or technology	The proposal was approved on a stated technical approach. Changing it changes the CCP claim, the risk profile and often the contribution itself.
Changing the deployment target or relaxing a non-functional target	Non-functional targets form part of the acceptance basis. They are not aspirations to be quietly lowered once they prove difficult.
Changing the evaluation baseline, metrics or success criteria	Adjusting the measure after seeing the result invalidates the evaluation. Such changes must be justified and made before the results are known.
Reassigning primary technical ownership between members	Individual assessment depends on traceable ownership. Reallocation is permitted but must be recorded when it happens rather than reconstructed at the end.
Adding scope beyond the committed core	Extension is welcome once the core is secure, but it must not place the committed scope at risk, and it should be recorded as an extension rather than absorbed.

Change is expected; undocumented change is not

A group that revises its approach on evidence and records the decision demonstrates engineering judgement and should present that revision as a finding. A group whose final system quietly differs

from its approved proposal, with no record of when or why it diverged, has lost the traceability on which its own report, its acceptance matrix and its individual assessment all depend.

Scope sufficiency questions

- Is the central problem complex enough to require analysis and design, rather than only routine implementation?
- Can the group identify meaningful technical decisions, risks and validation work across both semesters?
- Does each member have a defensible technical role with primary ownership and shared integration responsibility?
- Can the team demonstrate feasibility at the proposal defense, baseline requirements and design by Mid-1, reach a working core system by Final-1, and deliver an integrated, evaluated and deployed system by Final-2?
- Is there a clear minimum committed scope, with optional extensions separated from the core?
- Can the project be evaluated with evidence beyond a successful demonstration?

C.6. Can This Idea Be Specified?

An approved idea does not go straight to code. It becomes a scope of work, a proposal and initiation plan, a timeline of milestones and deliverables, a software requirements specification, a software design specification, a requirements traceability and acceptance matrix, and a test, deployment and handover plan. Together these form the project report, and they are the primary written evidence of the group's engineering work. An idea that cannot be written down in this form is not ready for approval, however appealing it sounds in conversation.

The specifiability test is straightforward. At proposal stage the group should already be able to sketch, in outline, what its requirements, design and acceptance criteria would look like. Precision is not expected this early and requirements will change; what is expected is that the group can name concrete functional requirements, state at least one measurable non-functional requirement, and describe the evidence that would prove a requirement has been met. Vagueness at this stage is not a documentation problem — it is a sign that the problem has not yet been analyzed.

Question to answer before approval	**Why it decides readiness**
Can you name at least five concrete functional requirements, each in the form "the system shall…"?	If requirements can only be expressed as themes — "smart", "AI-powered", "user-friendly", "automated" — the problem has not been analyzed yet.
Can you state at least one measurable non-functional requirement?	Performance, availability, accuracy, security, privacy, accessibility and deployment targets must be expressed as numbers or observable criteria, not as adjectives.
Can you name the actors and the main data the system will hold?	If the group cannot list who uses the system and what it stores, neither the design nor the privacy, security and ethics analysis can begin.
Can you state an acceptance criterion for your hardest requirement?	Every requirement must eventually trace to a design element and a test. A requirement that cannot be tested cannot be accepted and should not be committed to.

Question to answer before approval	Why it decides readiness
Can you draw a first architecture on one page?	Major components, interfaces, data flows and external dependencies should be sketchable at proposal stage, even though they will change.
Do you know which parts are most likely to change?	Naming the uncertain parts is a sign of understanding, not of weakness. A group that believes nothing will change has not yet thought hard enough about the problem.

A one-sentence test for vagueness

Complete this sentence for the hardest part of the project: "The system shall <do something specific> for <a named actor>, and we will know it works when <observable evidence>." If the group cannot complete it, the idea needs more work before it is taken to defense. If the group can complete it only for the easy parts, the scope is in the wrong place.

C.7. Will This Idea Survive Generative AI?

Generative AI has changed what routine implementation costs. Work that once justified two semesters — building a data-entry application, wiring an API, producing a conventional interface, generating boilerplate and scaffolding — can now be produced quickly with assistance. This does not make final year projects easier. It raises the threshold for what deserves capstone status. An idea whose principal difficulty was the volume of code required is no longer a sufficient idea.

The AI-resilience test is a thought experiment for choosing ideas, not a policy on tools. If a competent final-year student with good tooling could produce a credible working version of the proposed project in a few days, the idea is not yet an FYP. This is not a claim that such a version would be good. It is a signal that the project's difficulty never lay in the implementation. The group should then ask where the remaining difficulty actually is — in the data, the domain, the constraints, the trade-offs, the evaluation or the deployment — and make that the project. If there is no remaining difficulty, the group should choose a different problem.

Ask	What a defensible answer looks like
Could a strong student with AI assistance build a working version of this in a week?	If yes, identify precisely what such a version would get wrong — the data quality, the domain constraints, the edge cases, the failure behavior, the evaluation, the deployment reality — and make that the project. If nothing would go wrong, choose another idea.
What in this project cannot be generated?	Problem formulation, stakeholder study, data acquisition and curation, domain judgement, trade-off decisions, evaluation design, failure analysis, and deployment under real constraints. These are the parts that carry academic weight.
Where does assistance stop being helped and start being the project?	If the central capability is supplied by an external model or service, the contribution must lie elsewhere — in the data, the retrieval or pipeline design, the validation, the safeguards, the integration or the evaluation — and the group must be able to state what that is in one sentence.

Ask	What a defensible answer looks like
Can every member explain and modify code they did not write by hand?	Ownership is demonstrated by explanation, modification and defense under questioning, not by authorship of keystrokes. A member who cannot reason about a component does not own it and should not be credited with it.
Does the evaluation test the system, or only the happy path?	Assistance makes it easy to produce something that demonstrates well and fails quietly. Evaluation should target the cases where the system is most likely to be wrong.
Has the group followed current disclosure policy?	Permitted use of AI tools, generated code, prompts, datasets, models and text must be disclosed and acknowledged exactly as the university requires. Undisclosed use is an integrity matter, not a technical one.

AI raises the bar; it does not lower the standard

The permitted use of AI tools is governed by current university policy and is outside the scope of this guide. What is within scope is the consequence for idea selection: groups should choose problems whose difficulty lies in analysis, data, constraints, trade-offs, evaluation and deployment — the parts of engineering that assistance does not remove. Students remain fully responsible for the reasoning, design, implementation, verification and defense of everything they submit.

C.8. AI-Assisted Development in an FYP

Industry practice now describes software development along a maturity spectrum. At one end the lifecycle is entirely human-led. In the middle, humans lead both judgment and execution while AI assists with isolated tasks. At the far end — described as AI-driven or agentic development — AI leads execution and humans are repositioned as validators and strategists who approve generated designs, code and tests. The School's position is that a final year project is conducted at the middle level: AI-assisted development, not AI-led development.

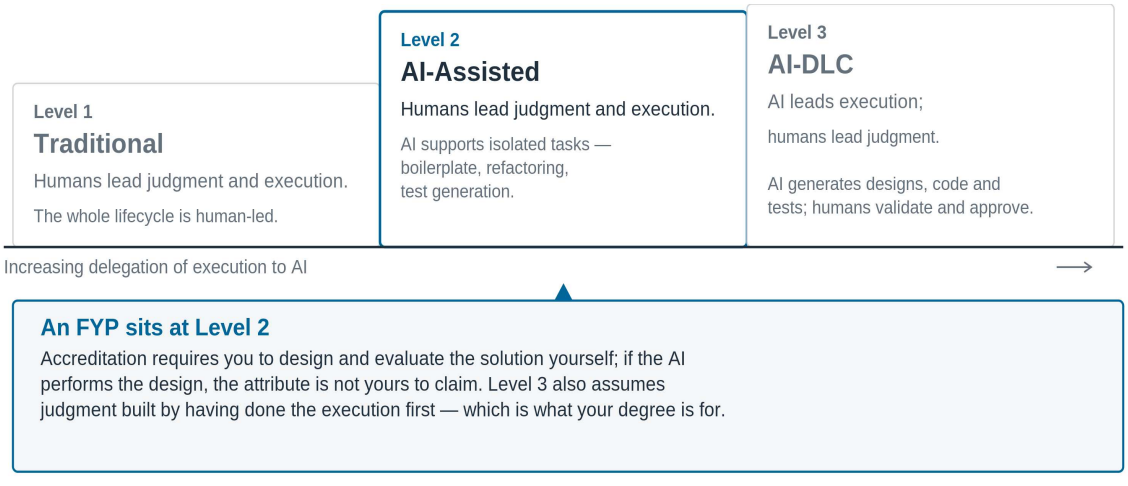


Figure 3. The three levels of development maturity. An FYP is conducted at Level 2.

C.8.1 Why an FYP is Conducted at the Assisted Level?

Reason	**Explanation**
Accreditation	Seoul Accord GA-4 requires the graduate to design and evaluate solutions for complex computing problems, and to design and evaluate systems, components or processes. Where an AI performs the design, the attribute has not been demonstrated by the graduate and cannot be reported as attained. GA-2, the abstraction and conceptualization of computing models, fails in the same way.
Judgment follows execution	AI-led development assigns people the validating role — approving designs, reviewing generated code, accepting recommended fixes. Those are senior roles, held by people who can validate because they have previously executed. A student validating work they could not have produced is not exercising judgment. Your degree is where the execution competence that makes judgment possible is built.
The ownership standard	Section 14 requires the member who owns CCP-relevant work to explain it, justify it against an alternative, predict what breaks and modify it under questioning. AI-led development is designed to relocate precisely that capability.
Infrastructure	AI-led development depends on an encoded knowledge base of business rules and architecture, tool integrations across an organization's systems, governance guardrails and production telemetry. None of that exists at project scale, and building it is not the FYP.

Assisted does not mean unassisted

This is not a restriction on using AI tools. Permission is governed by current university policy, use is expected, and permitted use that you disclose is not penalized. It is a statement about where responsibility sits. You may use assistance throughout the project. What you may not do is delegate the judgment your degree exists to develop or claim ownership of work you cannot account for.

C.8.2 What You Own and What AI may Assist

The division is not by tool and not by phase. It is by whether the work carries the reasoning a panel will assess. Anything that touches the complex computing problem — how it is formulated, how the solution is designed, which trade-offs were accepted, how it is evaluated and why it fails when it fails — remains yours regardless of what helped you produce it.

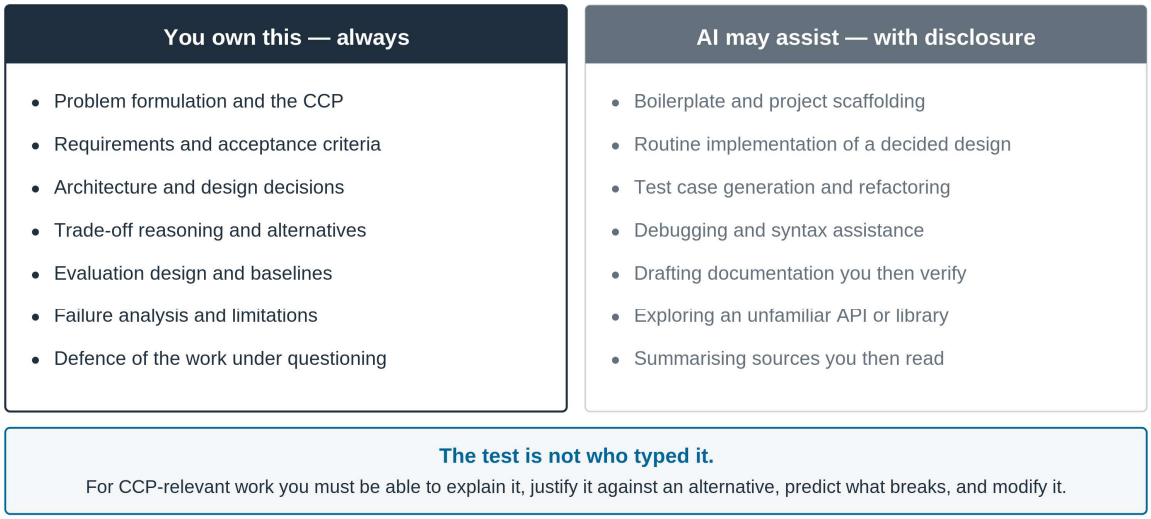


Figure 4. The ownership boundary. Assistance is judged by what you can account for, not by who produced the keystrokes.

Peripheral work is held to a lower standard: you should know what it does and why it is there, but you are not expected to defend every line of routine scaffolding. The distinction matters most at the boundary, so if you are unsure whether a component is CCP-relevant, assume it is and be ready to defend it.

C.8.3 Specification Before Generation

A consistent finding in AI-native practice is that generated output is only as good as the specification behind it: if the specification is vague, the output will be confidently wrong. A group that can state its requirements, constraints, acceptance criteria and interfaces precisely will get better results from any tool and will also produce the documentation the project report depends on. A group that cannot will generate plausible code for the wrong problem and will not notice.

Formulate the problem at requirement level first — what the system must achieve, which constraints cannot be violated, which objectives compete, and what happens on failure — before selecting a method or generating an implementation. Annex C shows what that formulation looks like at three levels of maturity.

C.8.4 Recording How AI is Used

Disclosure is not a formality, and it is not an admission. The School's position is that transparency is more useful than prohibition: a group that records where assistance was used can be assessed on what it understands, while a group that conceals it invites the assumption that it understands nothing. Permitted use that is disclosed is not penalized.

Maintain an AI usage log alongside your dependency disclosure and update it as you work rather than reconstructing it before a defense. It should record the activity, roughly where in the project it applied, and — for anything touching the complex computing problem — how the output was checked before you relied on it.

Activity	**What to record**
Ideation and problem exploration	Where assistance shaped the problem framing, the alternatives considered or the literature you went on to read yourself.
Coding assistance and generation	Which components, and whether they are peripheral or CCP-relevant. For CCP-relevant code, state how you verified it behaves as intended.

Activity	What to record
Debugging and diagnosis	Where assistance located or explained a fault, and whether you confirmed the diagnosis independently.
Test generation	Which tests were generated, and how you checked they test something real rather than restating the implementation.
Documentation and writing	Which artefacts were drafted with assistance and verified by you for technical accuracy before submission.
Anything else	Data cleaning, configuration, refactoring, translation, diagram drafting, or any other use you would not want a panel to discover unrecorded.

The log is evidence for you, not against you

Panels use the log to choose what to ask about, not to deduct marks for using tools. A group that records honest, specific entries and can explain the CCP-relevant ones demonstrate the ownership. An empty log on a project that plainly used assistance, or a log whose entries cannot be defended, is treated as a disclosure failure. The log is submitted with the dependency disclosure sheet and countersigned by your supervisor.

C.9. A Disciplined Idea-Discovery Process

Do not begin by selecting technology or asking for a list of trendy applications. Use the following sequence. Several ideas should be explored and rejected before one is taken to proposal defense.

Stream A groups usually enter this sequence at steps 2 and 4, because literature is where the idea begins but they must still complete steps 1 and 3 by identifying who is affected by the problem and confirming that the gap matters in practice. Stream B groups enter at step 1 and must not skip step 2, because the solution landscape is where most application ideas are discovered to be already solved.

Step	Purpose
1. Explore problem spaces	Identify users, current workflows, failures, consequences and constraints. Speak to relevant stakeholders where feasible.
2. Map the solution landscape	Study commercial products, open-source systems, research prototypes, standards, manual workflows and prior departmental FYPs.
3. Verify the gap	Use evidence to show what remains unsolved or inadequately solved. A missing local version is not automatically a technical gap.
4. Formulate the CCP	Identify which Seoul Accord characteristics genuinely apply, what trade-offs exist, and what knowledge/ analysis will be required.
5. Define the contribution	State what the project will add and how it differs structurally from the closest alternatives.
6. Compare technical approaches	Identify alternatives, expected advantages, risks, data/resource needs and the basis for selection.

Step	Purpose
7. Run a focused POC or technical spike	Test the highest-risk technical assumption before requesting approval.
8. Define evaluation and success criteria	Specify how correctness, usefulness, reliability or improvement will be measured.
9. Shape scope and ownership; also define DEI, ethical and legal issues	Separate core scope from optional work; allocate meaningful primary responsibilities and shared integration duties. Devise DEI and ethical and legal statements
10. Stress-test the idea	Ask what can fail, what may already exist, what depends on unavailable resources, and what evidence would cause the group to revise or reject the idea.
11. Check SDG/ thematic alignment	Identify one relevant SDG or thematic area and explain how it connects to the problem and stakeholders without manufacturing importance or complexity.

C.10. Product, Prior-FYP and Workflow Comparison

Comparison must be structural, not based only on titles or feature lists. Compare the problem, users, inputs, outputs, technical approach, constraints and evaluation. Claims about limitations should be supported by documentation, observed behavior, credible reviews, research evidence or stakeholder input where possible.

Existing solution/ prior FYP	Problem and target users	Core approach/ strong features	Evidence-based limitation	Implication for our project
Solution A				
Solution B				
Closest prior FYP				
Current manual/non-digital workflow				

Important: Changing the platform, language, user interface, model name or market does not by itself make a project different. The group should be able to explain how its problem formulation, constraints, pipeline, contribution or evaluation is materially different.

C.11. Research-to-Decision Table

Use research to make decisions, not to fill pages. Every important source should have a clear role in the project.

Source/ method/ standard	Main findings or method	Relevance and limitation	Decision influenced	Planned evidence
Source 1				

Source/ method/ standard	Main findings or method	Relevance and limitation	Decision influenced	Planned evidence
Source 2				
Source 3				
Source 4				

C.12. Proposal-Stage Proof of Concept (POC) or Technical Spike

Every proposal must include at least one focused feasibility proof, and it must be demonstrated at the proposal defense rather than promised for later. The proposal defense is the first formal evaluation stage of the FYP, and the POC is what separates a researched proposal from an attractive intention. It is not a polished MVP or a complete product but a high-fidelity functional prototype (a skeleton of the product) that tests the riskiest assumption. Its purpose is to establish that the greatest technical risk has been examined, and that the group has started investigating the problem rather than describing it.

A group that cannot show a POC at the proposal defense has not yet established feasibility, and its proposal should normally be deferred rather than approved. Where the highest risk is genuinely non-technical — access, permissions, ethical clearance, licensing or data availability — the evidence should address that risk instead, in the same disciplined form: what was assumed, what was tested, what was found, and what changes as a result.

The POC is a condition of approval, not a formality

The panel should ask to see the evidence, not a description of it. A slide stating that a POC was completed is not a POC. The group should be able to show what was run, on what input, with what result, and to say plainly what that result changed in the design, the scope or the plan. A POC that confirmed the assumption is useful; a POC that disproved it and caused the group to revise its approach is usually more valuable and should be presented as a finding rather than concealed as a setback.

Project risk	Suitable proposal-stage evidence
Data availability or quality	A sample acquisition/ cleaning exercise, annotation trial, data profile, privacy check or evidence that the required data can legally and practically be obtained.
Algorithm/ model feasibility	A baseline run, small reproduction, sample inference, comparison of candidate methods or analysis of errors on representative cases.
System or integration feasibility	A minimal end-to-end path, protocol/ API compatibility test, latency measurement, offline/ edge trial or failure/ fallback experiment.
Hardware/ IoT feasibility	Sensor reading, communication test, power/ range experiment, device integration or small physical prototype.
Security/ privacy feasibility	Threat model, access-control experiment, encryption/ performance test, data-flow analysis or proof that sensitive data can be handled safely.
Human/workflow feasibility	Clickable workflow, task walkthrough, early stakeholder feedback, accessibility check or usability risk test.

Project risk	Suitable proposal-stage evidence
Deployment feasibility	A test of the intended deployment target: installation in a clean environment, a container build, device provisioning, a platform or store constraint check, or a resource and cost estimate for the environment in which the system is meant to run.

A POC should answer five questions

1. What risky assumption was tested? 2. What input/ setup was used? 3. What result was obtained? 4. What was learned or disproved? 5. How will this change the design or plan? A polished interface that does not test technical or workflow risk is not sufficient by itself.

C.13. Plan Evaluation Before Approval

Testing, evaluation, and user acceptance are related but different, and a project needs all three. Testing asks whether the implemented system behaves as specified. Evaluation asks whether the proposed solution meaningfully addresses the problem, and how it compares with a baseline, existing approach, benchmark or stakeholder need. Acceptance asks whether the people the system is for will actually use it. This section covers evaluation planning; Section 13.1 sets out the distinction and the acceptance requirement.

Evaluation Question	Examples of Evidence
What is the baseline or comparator?	Existing product/ workflow, simple algorithm, previous method, manual process, published result or current practice.
What data, cases or users will be used?	Representative datasets, controlled scenarios, edge cases, user tasks, simulated workloads or field observations.
What metrics or criteria will be measured?	Accuracy, precision/ recall, latency, throughput, reliability, security findings, task completion, usability, cost, resource use or acceptance criteria.
What failure cases matter?	Incorrect outputs, unavailable services, noisy inputs, misuse, bias, unsafe recommendations, performance degradation or recovery behavior.
What conclusion can the evidence support?	Define the intended claim narrowly and state limitations. A successful demo alone supports only a limited conclusion.

C.13.1 Testing, Evaluation, and User Acceptance Are Three Different Questions

Groups routinely conflate these, present one, and believe they have covered all three. They answer different questions and need different evidence.

Question	What it asks	Typical evidence
Testing — does it behave as specified?	Whether the implemented system does what the requirements say, including at the edges and when things fail.	Unit, integration and system tests; edge and failure cases; defect record by severity.

Question	What it asks	Typical evidence
Evaluation — is it any good?	Whether the solution meaningfully addresses the problem, measured against a baseline, comparator, benchmark or stated success criteria.	Baseline comparison, metrics, ablations, failure analysis, stated limits of what the evidence supports.
Acceptance — will the people it is for actually use it?	Whether the intended users or stakeholders, working on realistic tasks, find the system usable, correct and worth adopting.	User acceptance testing with named participants, task outcomes, issues raised and resolved, and a recorded acceptance decision.

User acceptance testing is required for every project and is evidenced at FYP-2 Final; the plan for it must exist by FYP-2 Mid so that it is not attempted in the final week. Plan it at proposal stage: knowing who will accept the system usually changes what you build.

What your acceptance testing must record	Notes
Method	How testing was run — supervised task sessions, field trial, pilot deployment, structured walkthrough or expert review — and why that method suits the system.
Participants	How many, and who they are in relation to the intended users: real end users, operators, domain experts or proxies. State plainly which. Numbers are usually small; that is acceptable if the participants are the right ones and you do not complain about them.
Tasks and results	The tasks participants attempted, what succeeded, what failed, and any measures taken — completion, time, errors, assistance needed, satisfaction.
Issues found and what you did	Problems raised, which were fixed before final submission, which were deferred, and why. Issues found and fixed are evidence of a working process, not of failure.
Acceptance outcome	Whether the stakeholder accepts the system for its intended use, with any conditions or reservations recorded in their words rather than yours.
Recording	A short video of a representative session, with participant consent. It is evidence that the testing happened as described and lets examiners verify outside panel time.

Where real users cannot be reached

Some systems cannot be tested with their intended users inside a two-semester project — clinical tools, safety-critical systems, security tooling, or research artefacts with no direct operator. The requirement does not disappear; the participant definition changes. Use domain experts, operators, or the closest defensible proxy, declare the substitution at proposal stage rather than at the final defense, and state

clearly what an acceptance result from proxies can and cannot tell you. Consent, privacy and any institutional approval needed to involve participants are the group's responsibility and should be resolved early.

C.14. External Components and Student Ownership

Modern systems normally use frameworks, libraries, APIs, pretrained components, templates and other development resources. Their presence is neither a strength nor a weakness. What matters is what components supplies relative to the contribution the group is claiming.

Every important external component falls into one of three categories. The same three categories appear on the External Assistance and Dependency Disclosure Sheet in the evaluation forms, so a group that classifies its dependencies at idea stage will already have done most of the disclosure work it must submit later.

Category	What it means	What is required
Support	Tooling, infrastructure, scaffolding, boilerplate, standard libraries, UI kits, build and deployment tooling, and anything else that does not touch the claimed contribution.	Disclosure and license compliance. No further justification is expected, and panels will not spend time here.
Core-assist	A component that contributes to a part of the system relevant to the complex computing problem — a hosted model, a service API, a published implementation, or a template that shapes a central workflow.	Permitted. The group must justify the choice, state the component's limitations, validate its outputs, plan a fallback, and be able to explain and modify the surrounding design. The contribution must remain identifiable independently of the component.
Core-replacement	A component that supplies the contribution itself: the complex computing problem is solved by the component rather than by the group.	Not acceptable. Either the claimed contribution is redefined around what the group actually does, or the component is replaced by the group's own work. This is a scoping decision and should be taken before approval, not discovered at defense.

The boundary is the contribution, not the technology

A group may use a hosted model, a published implementation or a commercial API and still produce an excellent FYP — provided it can say in one sentence what remains its own. Apply the test in Section 4 directly: if the external component were replaced by a comparable alternative, would the claimed contribution survive? If the answer is no, the component is the contribution, and the project needs rescoping rather than defending.

C.14.1 The Standard of Understanding for CCP-Relevant Work

Panels do not ask who typed a line of code. They ask whether the member who claims a component can account for it. For any part of the system that touches the complex computing problem, the member responsible should be able to do four things.

The member responsible should be able to:	**What a panel will ask**
Explain what it does and why it is there	Walk through the component's role in the system and the sub-problem it addresses, without reading from the code.
Justify why it is built or configured this way	Name at least one alternative that was considered and say why it was rejected.
Predict what breaks	State what fails if an input, assumption, dependency, requirement or scale changes — and what the system does when it fails.
Modify it under questioning	Describe, or make, a specific change and identify what else must change with it.

A member who cannot do these four things does not own that component, whatever the repository history shows. The standard applies uniformly: it does not matter whether the code was written from scratch, adapted from a tutorial, taken from a published implementation, or produced with assistance. Peripheral code is held to a lower standard — the group should know what it does and why it is there but is not expected to defend every line of routine scaffolding.

Two further points. Ownership is individual. The obligation attaches to the member who claims responsibility, and the team should understand the interfaces between components even where only one member understands the internals.

Classification can change. If the group's complex computing problem claims shift during the project — which Section 5.3 permits when the change is recorded — a component previously classified as support may become core-assisted. The disclosure sheet should be updated when that happens, not reconstructed at the end.

Questions to ask about every important dependency:

- What exactly does the external component provide, and which part remains the students' technical responsibility?
- Would the proposed FYP still contain a meaningful computing contribution if that component was replaced?
- How will output, failures, limitations, cost, security and fallback behavior be tested?
- Can the group explain and modify the surrounding design and integration?
- Has the group correctly cited or attributed external code, data, models, prompts, text and documentation where required?

C.15. Common Weak Idea Patterns

Weak pattern	**Why it is weak**	**What would make it defensible**
Routine portal, CRUD system or feature-rich application	Size and screen count do not establish a CCP or contribution.	Identify a genuine technical/ workflow problem, design trade-offs and evaluation beyond implementation.

Weak pattern	Why it is weak	What would make it defensible
Generic chatbot or retrieval wrapper	Central capability may be provided by an external model/service.	Define a domain problem, retrieval/data design, validation, safeguards, failure analysis and contribution beyond the service.
Model training with no problem or baseline	A model and an accuracy value do not show useful problem solving.	Define the use case, dataset quality, baseline, metrics, practical constraints, error analysis and system integration.
Local copy of an existing product	Localization alone is usually not a computing contribution.	Show a local constraint that changes requirements, architecture, data, method or evaluation.
Implementation of a paper without understanding	Reproduction can be valuable, but copying steps is not enough.	Explain the method, reproduce transparently, compare results, analyze differences and adapt or integrate it meaningfully.
Independent member modules with weak integration	The project lacks coherence and shared system responsibility.	Use one central problem, defined interfaces, integrated evaluation and shared architecture/ quality responsibilities.
Industry feature delivery only	Client satisfaction does not by itself show academic learning.	Map the work to problem analysis, design decisions, testing, evaluation and individual technical ownership.
SDG/ thematic label without problem connection	A broad theme or SDG label does not prove technical depth, stakeholder need or contribution.	Explain the real problem and stakeholders first, then show a natural and limited SDG/ thematic connection.
Project whose only difficulty is the volume of code	Scale of implementation no longer establishes capstone-level difficulty, because routine code is now cheap to produce.	Relocate the difficulty into problem formulation, data, constraints, trade-offs, evaluation or deployment, and state where it now sits.
Paper reproduced with no system (Stream A)	The result in a notebook is not an FYP artefact; the standard requires a system that can be deployed and operated.	Engineer the method into an operable system with an identified user, a stated deployment target and an evaluation on realistic input.
Application built with no research basis (Stream B)	A working product does not by itself demonstrate research-informed problem solving, a CCP or a defensible contribution.	Establish the evidence base, identify the complex computing problem in the constraints, and define the evaluation before adding features.
System that runs only on the developer's machine	Deployability is part of the core standard; a system that cannot	Define the deployment target early and test installation, configuration

Weak pattern	Why it is weak	What would make it defensible
	be installed and operated by anyone else is incomplete.	and operation by someone outside the team.

C.16. Proposal Defense Package

The proposal presentation should be concise and should present evidence before the panel begins questioning. The slide order may be adjusted, but all of the following information should be visible.

Suggested slide	Required content
1. Title and one-line problem	Project title, members, supervisor, declared stream (research-derived or application-derived) and a one-sentence problem statement.
2. Stakeholders and current workflow	Who faces the problem, how it is currently handled, why the failure matters, and the one relevant SDG or thematic area that gives broader context.
3. Existing solutions and prior FYPs	Structural comparison and evidence-based gap.
4. Research and technical basis	Relevant current/ authoritative work, alternatives studied, and decisions influenced.
5. CCP/ complex activity justification	Applicable Seoul Accord characteristics, central trade-offs, uncertainty and why routine development is insufficient.
6. Proposed contribution and solution	Contribution statement, high-level architecture, major components, boundaries and the intended deployment target.
7. POC/ technical spike (mandatory)	Using a high-fidelity functional prototype, the risk that was tested, the evidence obtained — shown, not described — what was learned or disproved, and the resulting design or plan decision. A proposal that cannot demonstrate feasibility evidence should not normally be approved.
8. Evaluation plan	Baseline, data or scenarios, metrics, success criteria, at least one measurable non-functional criterion, and the failure cases that matter most.
9. Scope, risks and dependencies	Committed scope, optional extensions, resources, data, modern tools and external dependencies, professional and ethical compliance, deployment and integration risk, and the fallback plan.
10. Team responsibilities and iterations, DEI and ethical/ legal issues	Primary technical ownership, shared integration/ quality work and FYP-1/ FYP-2 milestones, DEI statement, ethical and legal statement

C.17. Final Idea-Readiness Checklist

[ ] We can state the problem, users, current workflow, gap and contribution in one minute.
[ ] We have identified one relevant SDG or thematic area and can explain its connection without using it as artificial complexity.

[ ] Our gap is supported by comparison with products, open-source systems, prior FYPs, research or current practice.
[ ] Our research/technical sources influence specific design or evaluation decisions.
[ ] We have identified the central CCP and can justify the applicable Seoul Accord characteristics.
[ ] The complexity arises from the real problem, not from adding technologies or features.
[ ] Our contribution is meaningful even if it is reproduction, adaptation, evaluation or constrained system design rather than new algorithmic novelty.
[ ] We have a focused proposal-stage POC or technical spike that tests the assumption carrying the greatest risk, and we can demonstrate the actual evidence at the defense rather than describe it.
[ ] We can state what our POC confirmed or disproved, and exactly what it changed in our design, scope or plan.
[ ] If we are implementing or improving a state-of-the-art method, we can explain its mechanism, assumptions and failure modes without the code in front of us, and we can state our baseline and our delta.
[ ] We have defined a baseline, success criteria, evaluation evidence and important failure cases.
[ ] Every important external component is classified as support, core-assist or core-replacement, and nothing we claim as our contribution sits in the core-replacement category.
[ ] For every part of the system that touches our CCP, the responsible member can explain it, justify it against an alternative, predict what breaks if it changes, and modify it under questioning.
[ ] We have disclosed and acknowledged permitted external code, AI tools, licensed data, copyrighted material, datasets, models, APIs and templates according to current policy.
[ ] The project is one coherent system with meaningful primary technical ownership for all members and shared integration responsibility.
[ ] We have separated committed scope from optional extensions and can explain what should be complete in FYP-1 and FYP-2.
[ ] We know who will accept this system, how acceptance will be tested, and whether they are real users or declared proxies.
[ ] We can state where the system will be deployed for real, and what would prevent that deployment if anything would.
[ ] We are keeping an AI usage log as we work, and can defend every CCP-relevant entry in it.
[ ] Every member can explain the complete idea and defend at least one important technical decision.
[ ] We understand data, legal, ethical, security, privacy, cost, deployment and dependency risks relevant to the project.
[ ] We have checked structural overlap with previous departmental FYPs, not only title similarity.
[ ] We can answer: what will still require serious analysis, engineering and evaluation after routine implementation work is complete?
[ ] We have declared our stream and can explain whether the idea entered through published research or through an observed real-world problem.
[ ] We can state where the system will run when deployed, and who besides us could install and operate it from our documentation.
[ ] We can name concrete functional requirements, at least one measurable non-functional requirement, and an acceptance criterion for our hardest requirement.
[ ] We have asked whether a competent student with AI assistance could produce a credible version of this quickly, and we can say where the remaining difficulty lies.

Final decision rule

Do not select an idea because it is fashionable, easy to demonstrate or rich in features. Select it when the group can defend the problem, research basis, complex computing challenge, contribution, feasibility, evaluation and ownership as one coherent story.

C.18 — Stream A: Starting from Published Research

Stream A projects begin from published work: a method, a benchmark, a dataset or an open problem stated in the literature. The strength of this entry point is that the technical foundation is already articulated, and the evaluation vocabulary already exists. Its characteristic weakness is that a group can spend two semesters re-running someone else's code and arrive with neither a system nor an understanding. This annex addresses that risk. It supplements the common requirements in Sections 1 to 17 and does not replace any of them.

C.18.1 Choosing the Research Paper

The choice of paper determines most of the project's risk. Make it deliberately and test the riskiest assumption before requesting approval — not in the fifth month.

Criterion	**What to check before committing**
Standing and recency	Prefer current peer-reviewed work from credible venues, supplemented by seminal papers and official standards where the foundations matter. A single unreviewed preprint with no independent validation is a weak basis for two semesters of work.
Reproducibility	Are the method, parameters, data and evaluation protocol described well enough to reproduce? Is code or data released? If not, the reproduction itself becomes the principal risk and must be tested at proposal stage.
Resource feasibility	Can the method be trained, fine-tuned or run within the compute, storage, license, budget and time actually available to the group? Verify this in the proposal-stage spike and identify a fallback method before approval.
Data availability	Is the dataset obtainable, legally usable and representative of the intended deployment context? Data that the group must collect or annotate is a project in itself and must be scoped, resourced and ethically cleared as such.
Room for contribution	Does the paper leave a stated limitation, an untested setting, a missing comparison, an unexamined assumption or an unaddressed constraint? A paper with no visible opening leaves the group with nothing to add.
Path to a system	Can the method plausibly become something a user or operator can run? If the answer is no, the idea fails the deployability requirement in Section 1.2 regardless of its research merit.

C.18.2 Establishing Your Delta

The delta is what the project adds beyond the paper. It need not be novelty in the sense of invention. A rigorous reproduction, a well-argued adaptation, a fair comparison or serious engineering under constraints are all legitimate — provided the group states clearly what it is doing and evaluates it honestly.

Type of delta	What must be shown
Faithful reproduction with transparent reporting	Where published results are difficult to reproduce, careful reproduction is a legitimate contribution — provided the group reports the protocol, every difference from the original, the results obtained, and a reasoned analysis of any divergence.
Adaptation to a new context	A different domain, language, population, sensor, data regime or regulatory setting — with evidence that the change is non-trivial and an analysis of what it does to performance and to the method's assumptions.
Constraint-driven engineering	Running the method under a constraint the paper did not consider: latency, memory, offline operation, edge hardware, cost, energy, privacy or continuous operation. State the constraint as a number, not as an aspiration.
Comparison and benchmarking	A fair, documented comparison of candidate methods on a problem or dataset where no such comparison exists, with an honest account of what the comparison can and cannot support.
Failure and limitation analysis	Systematic characterization of where the method breaks, on which inputs and why — particularly valuable where the original evaluation was narrow or the reported setting was favorable.
Integration into an operable system	Engineering the method into a deployable system with real inputs, real users, real latency and real failure handling. In most Stream A projects this is the largest and most defensible contribution.

State the delta in one sentence

"Paper X reports result A on dataset B under assumption C. We will reproduce, adapt or extend it to D, where assumption C does not hold, and evaluate using E." If the group cannot complete this sentence, the paper has not yet been understood well enough to build on. If the sentence is true but trivial, the delta is too small for two semesters.

C.18.3 From a Result to a System

A reproduction that lives in a notebook does not satisfy the School's standard. Stream A groups must plan from the beginning how the method becomes a system: what the inputs are in real use, who supplies them, what happens when the model is wrong, slow or unavailable, how the system is deployed and monitored, and what an operator actually sees. This engineering work is not a lesser activity appended to the research. In most Stream A projects it is precisely where the complex computing problem lives — in the trade-offs between accuracy and latency, capability and cost, automation and human oversight, and in the gap between a curated benchmark and messy real input.

C.18.4 Common Stream A Hazards

Hazard	How to avoid it
Running released code without understanding it	Require each member to explain the method's core mechanism, its assumptions and its failure modes without reference to the code. If they cannot, the group has not yet started the project.
Choosing a paper that cannot be run on available resources	Test the heaviest resource assumption in the proposal-stage spike, and have a documented fallback method identified before approval is requested.
Selecting a method the group cannot explain	Working with recent, high-performing methods is encouraged, and implementing or improving one is a strong basis for an FYP. Selecting one the group cannot explain is not. A well-understood recent method is the strongest position; a well-understood older method is a better project than a poorly understood recent one.
Treating the metric as the goal	An improvement in a benchmark figure is not automatically an improvement for a user. Connect every metric to the stakeholder consequence it is standing in for.
Benchmark data that does not resemble real input	Test the method early on data that looks like what the deployed system will receive. Degradation in realistic input is a finding worth reporting, not a failure to hide.
Leaving the system to the last month	Plan the end-to-end path early, even crudely, so that integration and deployment risk surfaces in FYP-1 rather than in the closing weeks of FYP-2.

C.19 — Stream B: Starting from a Real-World Problem

Stream B projects begin from something observed: a workflow that fails, a task done manually at unnecessary cost, a need that no available product serves well. The strength of this entry point is that the problem is real and the users exist. Its characteristic weakness is that a group builds a competent application around a problem that never required computing depth and then finds at defense that it cannot identify a complex computing problem. This annex addresses that risk. It supplements the common requirements in Sections 1 to 17 and does not replace any of them.

C.19.1 Validating that the Problem is Real

A problem is not established by assertion. Before an idea is defended, the group should have evidence that the problem exists, that it matters, and that it is not already adequately solved.

Evidence	What counts
Stakeholder contact	Conversations, observation, interviews or a small, structured survey with people who actually face the problem. Assumed users are not stakeholders. Record who was consulted, when, and what was learned.

Evidence	What counts
The current workflow	A concrete description of how the task is done today, including its cost in time, money, error or risk. If the current workflow cannot be described, any improvement over it cannot be measured.
Consequence of failure	What goes wrong when the current approach fails, and to whom. This determines how seriously the proposed system must be tested, safeguarded and evaluated.
Why existing products do not serve it	A structural comparison against real products, not a dismissal by feature list. “It is not available here” and “it is expensive” are market observations rather than technical gaps — though either may create a genuine constraint worth engineering around.
Constraints that are not negotiable	Connectivity, cost, device capability, literacy, language, regulation, privacy, or existing systems that must be integrated with and cannot be changed. These constraints are usually where the computing complexity comes from.

C.19.2 Finding the Complex Computing Problem

The most common Stream B failure is a good problem with a shallow solution. The remedy is not to add technology; adding technology produces artificial complexity, which Section 3 explicitly warns against. The remedy is to look harder at the constraints. Complexity in application-derived projects rarely comes from the features. It comes from what the real environment refuses to allow.

Source of genuine complexity in application projects	What it looks like
Conflicting stakeholder requirements	The regulator, the operator and the end user want incompatible things, and the design must resolve the conflict explicitly rather than quietly favor one party.
Hard environmental constraints	Intermittent connectivity, low-end devices, limited power, high-latency links, offline-first operation, or integration with a system whose behavior cannot be changed.
Data that is messy, scarce, sensitive or contested	Incomplete records, inconsistent formats, personal or regulated data, multiple sources that disagree, or data that must be acquired and curated before anything can be built.
Correctness with real consequences	Decisions affecting health, safety, money, legal standing or access to a service — requiring validation, auditability, human oversight and defensible failure behavior.
Scale or timing behavior	Concurrency, real-time deadlines, large volumes, or workloads that make the straightforward implementation unusable in practice.
Interoperability	Systems, protocols, standards or organizations that must be made to work together where no single party controls the whole.

Source of genuine complexity in application projects	What it looks like
Trust, security and privacy	Adversarial users, sensitive data, access control across roles or organizations, or a requirement to prove afterwards what the system did and why.

If the constraints disappeared, would the problem disappear?

Ask: if connectivity were perfect, the data were clean, every user was an expert and failure had no consequence — would this still be a hard computing problem? If not, then the constraints are the project, and they must be stated, designed for and evaluated explicitly rather than mentioned in passing. If nothing hard remains under any conditions, the idea is not yet an FYP.

C.19.3 What Stream B Groups Most Often Need to Add

Requirement	What is usually missing
Research basis	A study of comparable products, research literature, standards and prior FYPs that visibly changes the design — not a literature chapter written after the system was built.
A measured baseline	The current manual or existing digital workflow, measured. Without it there is nothing to compare against and no way to demonstrate improvement.
Evaluation beyond demonstration	Task-completion studies, measured performance against the baseline, error and failure analysis, or stakeholder evaluation on real tasks. A working demonstration supports only a very limited conclusion.
Explicit non-functional targets	Latency, availability, security posture, accessibility standard, offline behavior or data protection, stated as measurable criteria and tested.
Technical depth in at least one subsystem	At least one component whose design required genuine analysis and trade-off, which the group can defend in detail under questioning.

C.19.4 Common Stream B Hazards

Hazard	How to avoid it
Feature accumulation	Adding modules, roles and dashboards to create the appearance of scope. Commit to a narrow core and deepen it; move everything else to clearly separate optional extensions.
Assumed users	Building for an imagined stakeholder. Speak to real ones early, and record what changed in the requirements as a result of doing so.

Hazard	How to avoid it
The “local version” trap	Rebuilding an existing product for a local market. This is defensible only when a local constraint genuinely changes the requirements, architecture, data or evaluation — and the group must say which.
Interface treated as contribution	A usable interface is expected, not claimed as a contribution. The contribution must lie in the problem, the design, the data or the evaluation.
Ethics and data access discovered late	Consent, permissions, personal data handling and institutional approval should be resolved at proposal stage. Projects that depend on access nobody has agreed to are high-risk by construction.
Late discovery that there is no complexity	Identify and state the complex computing problem in the proposal, not after the application has been built. If it cannot be found at proposal stage, it will usually not appear later.

C.20 — Three-Level CCP Formulation Examples

The hardest part of idea selection is not understanding the definition of a complex computing problem; it is recognizing the difference between a formulation that sounds ambitious and one that actually is. This annex shows twelve domains, each formulated at three levels of maturity. Reading the contrast is usually more useful than re-reading the characteristic table in Section 3.

Examples C.1 to C.6 are application-derived (Stream B). Examples C.7 to C.12 are research-derived (Stream A). The domains are illustrative only: they are not a menu of approved projects, and a group that adopts one of these problems still has to establish its own evidence, stakeholders, gap and contribution.

The three levels are not arbitrary — they follow the Seoul Accord

Seoul Accord Section D.4.1 defines three tiers of computing problem. A Well-defined problem “can be solved in standardized ways” and is “encompassed by standards and/or documented procedures of practice”. A Broadly-defined problem “can be solved by application of well-proven analysis techniques” and “belongs to families of familiar problems”. A Complex problem “has no obvious solution, and requires conceptual thinking and innovative analysis”, and “is outside problems encompassed by standards and standard practice”. The Weak, Developing and Strong levels below correspond to those three tiers. This matters: The Accord scopes Well-defined problems to the computing technician and Broadly-defined problems to the computing technologist and reserves Complex problems for the computing professional — the graduate an accredited BS program produces. A Developing formulation is therefore not simply a less ambitious FYP; it sits below the level the accreditation expects of your degree.

The Strong formulations share a structure worth noticing. Each states the required outcome, the constraints that cannot be violated, the objectives that compete, the uncertainty the system must survive and the consequence of getting it wrong — and each does so before naming any method, model, framework or architecture. Formulating the problem at requirement level first is what makes the difficulty visible. A group that begins by choosing a technology has usually already decided the answer and will struggle to explain what made the question hard.

Application-derived examples (Stream B)

C.20.1 University Timetabling Product

Level	**CCP/ problem statement**
Weak	Develop an automated university timetable system that allocates courses, faculty members and rooms while avoiding clashes.
Developing	Develop a timetable-generation system that considers faculty availability, room capacities, laboratory requirements, student course clashes, instructor preferences and preferred teaching hours.
Strong	Develop a timetable-generation and repair system that must satisfy zero faculty, room and student-batch clashes; laboratory, equipment, accessibility and restricted-availability constraints; and must balance compact student schedules, fair distribution of undesirable hours, instructor preferences, room utilization and minimum disruption when late changes make all preferences impossible to satisfy simultaneously.

C.20.2 Institutional Information Assistant

Level	**CCP/ problem statement**
Weak	Develop a RAG-based university chatbot that answers questions from university documents.
Developing	Develop a university assistant that processes policies and handbooks, provides citations, supports multiple user roles and updates its knowledge base when documents change.
Strong	Develop a university information assistant that must answer from documents differing in authority, effective date and access level; distinguish current from superseded rules; prevent restricted-information disclosure; identify conflicting or insufficient evidence; and balance answer completeness, response time and the risk of unsupported responses.

C.20.3 Fraud Investigation Product

Level	**CCP/ problem statement**
Weak	Develop an AI-based system that detects fraudulent transactions and displays alerts on a dashboard.
Developing	Develop a fraud-monitoring product that combines transaction, account, device and channel history to calculate risk scores and prioritize alerts.
Strong	Develop a fraud-investigation product that must identify coordinated and evolving behavior across accounts, devices, channels and time while controlling false positives, preserving privacy, producing traceable explanations, prioritizing limited investigator capacity and meeting near-real-time response constraints.

C.20.4 Coordinated Traffic-Control Product

Level	CCP/ problem statement
Weak	Develop a smart traffic system that detects vehicles and changes signal timings.
Developing	Develop a traffic-management system for multiple intersections that consider vehicle volume, pedestrians and emergency vehicles.
Strong	Develop a coordinated traffic-control product that must preserve pedestrian and intersection safety, prioritize emergency vehicles, balance waiting time across competing traffic streams, prevent local optimization from worsening network congestion and continue operating under uncertain flows, sensor errors and communication delays.

C.20.5 Recruitment Decision-Support Product

Level	CCP/ problem statement
Weak	Develop an AI-based recruitment platform that ranks CVs against job descriptions.
Developing	Develop a recruitment product that evaluates education, experience and skills, ranks applicants and provides explanations to recruiters.
Strong	Develop a recruitment decision-support product that must reason from incomplete and inconsistently presented evidence, balance conflicting job criteria, control unfair exclusion and proxy bias, preserve auditability, support justified human overrides and communicate uncertainty where candidate evidence is insufficient.

C.20.6 Environmental-Risk Monitoring Product

Level	CCP/ problem statement
Weak	Develop an IoT system that displays sensor readings and sends alerts when values cross thresholds.
Developing	Develop an environmental monitoring product that combines readings from multiple sensors, reports trends and detects abnormal conditions.
Strong	Develop an environmental-risk monitoring product that must distinguish hazardous events from sensor drift, failure and communication loss; combine heterogeneous observations of different reliability; preserve timely alerts under power and bandwidth limits; and balance missed-event risk against false alarms that cause unnecessary intervention.

Research-Derived Examples (Stream A)

In a research-derived project the published work normally provides the starting point. The Strong formulation must still identify the substantial problem that remains the group's own, and the operating requirements of the product the contribution will sit inside.

C.20.7 Diagnostic-Support Product (Medical Imaging)

Level	CCP/ problem statement
Weak	Reproduce a published deep-learning model for medical-image classification.
Developing	Apply and compare published models on a local dataset and improve classification accuracy.
Strong	Develop a diagnostic-support product that must operate on heterogeneous, low-quality and partially missing field images; recognize distribution shift and uncertain cases; control clinically dangerous false negatives; provide traceable evidence; and operate within the compute and connectivity limits of the intended setting.

C.20.8 Urdu Opinion-Analysis Product

Level	CCP/ problem statement
Weak	Fine-tune a published language model for Urdu sentiment analysis.
Developing	Develop an Urdu sentiment model that handles Roman Urdu, code-switching and spelling variation.
Strong	Develop an Urdu opinion-analysis product that must handle code-switching, Roman Urdu, sarcasm, evolving vocabulary, scarce and inconsistent labels, domain shift and unequal performance across user groups while providing interpretable evidence for high-impact moderation or decision-support cases.

C.20.9 Federated Learning Product

Level	CCP/ problem statement
Weak	Implement federated learning using an existing framework and default aggregation.
Developing	Train a shared model across multiple organizations without exchanging raw data and compare it with centralized training.
Strong	Develop a federated learning product that must maintain useful and fair performance across highly non-identical institutional data, intermittent participation, unequal client resources, communication limits, privacy threats and unreliable client updates without exposing local data.

C.20.10 Misleading-Chart Review Product

Level	CCP / problem statement
Weak	Implement a published model for detecting misleading charts.
Developing	Detect multiple chart types and classify common misleading visual patterns.

Level	CCP / problem statement
Strong	Develop a chart-review product that must analyze charts extracted from heterogeneous images, PDFs and dashboards; reason with incomplete metadata and surrounding context; distinguish design error from intentional distortion; control costly false accusations; explain supporting evidence; and route uncertain cases for human review.

C.20.11 Legal Information Product

Level	CCP/ problem statement
Weak	Fine-tune a pre-trained language model for legal question answering.
Developing	Develop a legal question-answering system that retrieves jurisdiction-specific documents and provides citations.
Strong	Develop a legal information product that must distinguish binding, advisory, superseded and jurisdiction-specific sources; handle conflicting authorities and changing law; prevent restricted-document disclosure; identify insufficient evidence; and balance answer usefulness with the consequences of unsupported guidance.

C.20.12 Operational Intrusion-Analysis Product

Level	CCP/ problem statement
Weak	Reproduce a published intrusion-detection model on a public benchmark dataset.
Developing	Compare several intrusion-detection models and improve detection accuracy on local network data.
Strong	Develop an operational intrusion-analysis product that must detect evolving and low-frequency attacks from imbalanced, partially labelled and changing traffic; control alert overload; preserve response time; remain robust to adversarial manipulation; provide evidence suitable for investigation; and adapt without forgetting previously detected behaviors.

C.20.13 Which Characteristics each Example Exhibits?

The numbers below refer to the nine Seoul Accord characteristics listed in Section 3. Every Strong formulation exhibits at least six of the nine, and all twelve exhibit characteristics 1, 2, 3 and 5 — including the two that most clearly separate a complex problem from a routine one: no obvious solution, and beyond standard practice. None exhibits all nine by default, which is the point: the Accord requires “some or all”, not a full set.

Example (Strong formulation)	Characteristics clearly exhibited	Present only weakly
C.1 University timetabling	1, 2, 3, 5, 6, 8	4, 7, 9
C.2 Institutional information assistant	1, 2, 3, 4, 5, 6, 8, 9	7
C.3 Fraud investigation	1, 2, 3, 4, 5, 6, 7, 8, 9	—

Example (Strong formulation)	Characteristics clearly exhibited	Present only weakly
C.4 Coordinated traffic control	1, 2, 3, 4, 5, 6, 7, 8	9
C.5 Recruitment decision support	1, 2, 3, 4, 5, 6, 7, 8, 9	—
C.6 Environmental-risk monitoring	1, 2, 3, 4, 5, 7, 8, 9	6
C.7 Diagnostic support (imaging)	1, 2, 3, 4, 5, 7, 8, 9	6
C.8 Urdu opinion analysis	1, 2, 3, 4, 5, 7, 9	6, 8
C.9 Federated learning	1, 2, 3, 4, 5, 6, 8	7, 9
C.10 Misleading-chart review	1, 2, 3, 4, 5, 7, 9	6, 8
C.11 Legal information	1, 2, 3, 4, 5, 6, 7, 8, 9	—
C.12 Operational intrusion analysis	1, 2, 3, 4, 5, 7, 8, 9	6

Read the third column as carefully as the second

Characteristic 6, diverse stakeholders, is clearly present in only seven of the twelve examples — the most commonly missing characteristic in the set. Technically strong formulations often understate who else has a claim on the system's behavior, and panels should probe it. A group whose problem genuinely has one stakeholder group should say so plainly rather than invent others; a group that has simply not looked should look again before defending.

How to Use These Examples

Dos	DON'Ts
Use the contrast diagnostically: write your own formulation, then decide honestly which of the three columns it most resembles.	Do not adopt a domain from this annex because it appears here. These are illustrations of formulation quality, not pre-approved project titles.
Notice that every Strong statement names competing objectives and a failure consequence. If yours names neither, it is probably a Developing formulation.	Do not manufacture constraints to reach the Strong column.
Formulate at requirement level first, and let the method, architecture and technology follow the constraints you identified.	Do not treat the Strong statement as a scope commitment. It states the problem; Section 5 governs how much of it a group commits to across two semesters.
Expect the panel to ask which characteristics apply to your project and what project-specific evidence supports each claim.	Do not claim characteristics you cannot evidence. A short, honest list of four well-supported characteristics defends better than nine asserted ones.

Sources

- Seoul Accord, Section D: Graduate Attributes, especially D.4 (range of problem solving and computing activities) and D.5 (problem analysis, design/development, modern tool usage, teamwork and communication).
- National Computing Education Accreditation Council (NCEAC), Accreditation Manual v2.1, especially Sections 3.2.2 and 3.2.3.5. The manual describes FYDP as a two-semester capstone mainly involving literature search, individual analysis, formulation of a complex problem and its solution through complex computing activity.
- Higher Education Commission (HEC), Revised Curriculum for BS Computer Science (2025), especially Annex-B: Final Year Project Guidelines for Computing Programs. The guideline requires alignment with relevant SDGs or thematic areas, clearly defined student roles, stage-wise evaluation through proposal, progress reviews, final demonstration and report submission, appropriate final deliverables, and compliance with ethical, plagiarism, licensing, copyright and AI-tool acknowledgement requirements.
- FAST School of Computing's existing FYP process, proposal defense SOP, evaluation forms and practices, together with feedback from faculty supervisors, evaluators, the FYP committee and academic leadership during the review of the FYP process.
- The School of Computing's project documentation standard — scope of work, proposal and initiation plan, timeline of milestones and deliverables, software requirements specification, software design specification, requirements traceability and acceptance matrix, and test, deployment and handover plan — together with its five approval gates, which define the written deliverables that follow idea approval and together form the project report.
- Industry practice on AI-assisted and AI-driven development lifecycles, including the AI-Driven Development Life Cycle (AI-DLC) described by Amazon Web Services and related commentary on development-maturity levels. These sources inform the maturity spectrum in Section 8. They describe commercial practice rather than accreditation requirements, and where they conflict with Seoul Accord graduate-attribute obligations, the latter govern.